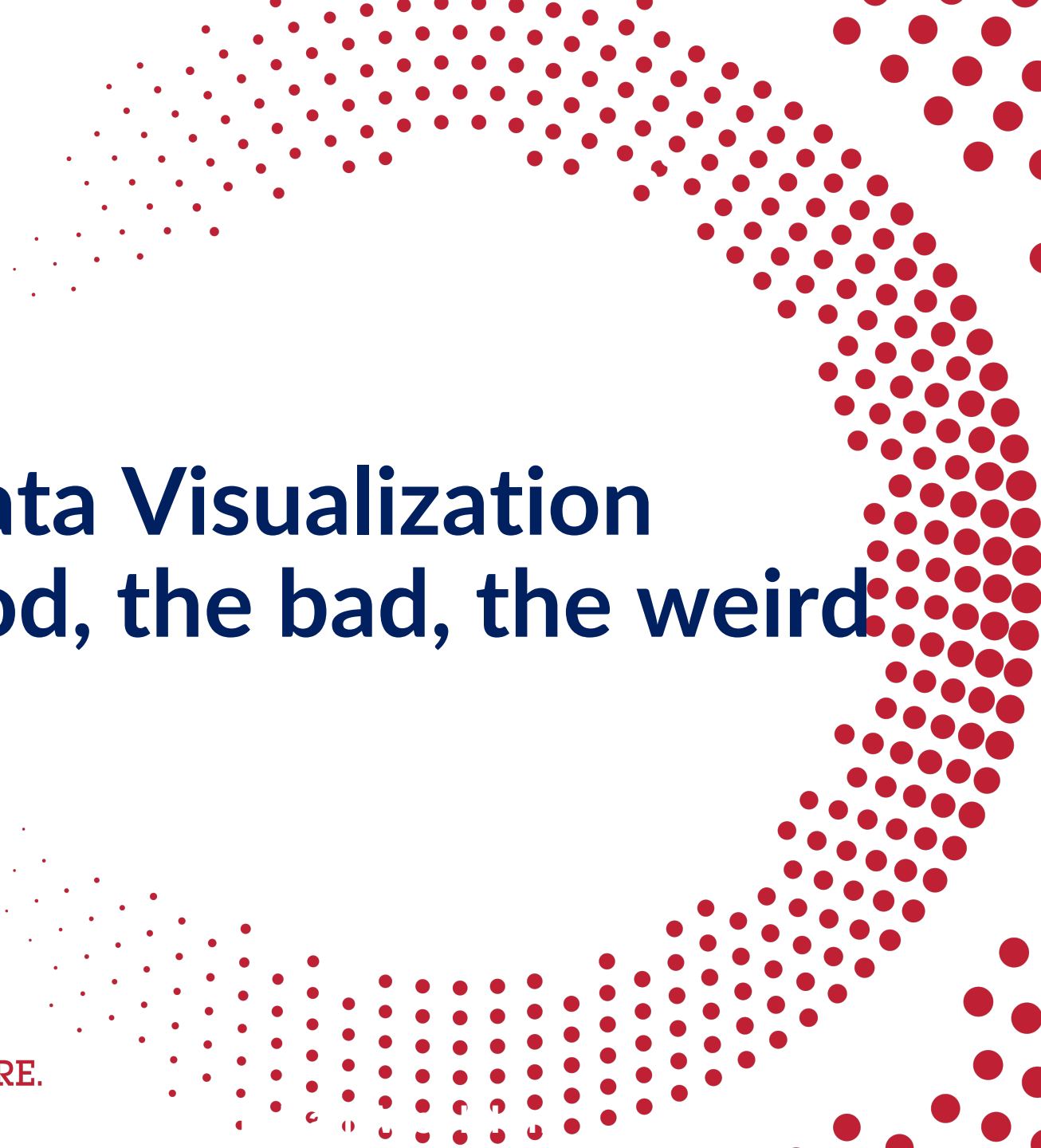




Data Visualization

The good, the bad, the weird

ONE LOVE. ONE FUTURE.

Goal

- Rules of thumbs to critique visualization design

Activity

- Create at least three sketches to visualize these 2 quantities

42

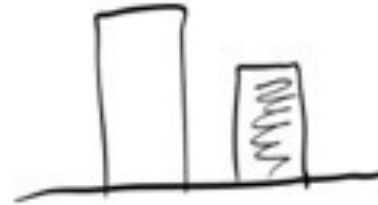
23

Most likely results

Pie Chart



Bar Chart



Scatterplot



23

42

Arabic Numbers

There are numerous ways to draw even
just
two numbers...

Which one is the best and why?

It depends on

- Data types
 - e.g., table, network, spatial, temporal
- Context of the data
- Tasks to perform
 - e.g., identify trends, compare values
- Questions to answer
- Messages to deliver

But, is there at least a guide for visualization design?

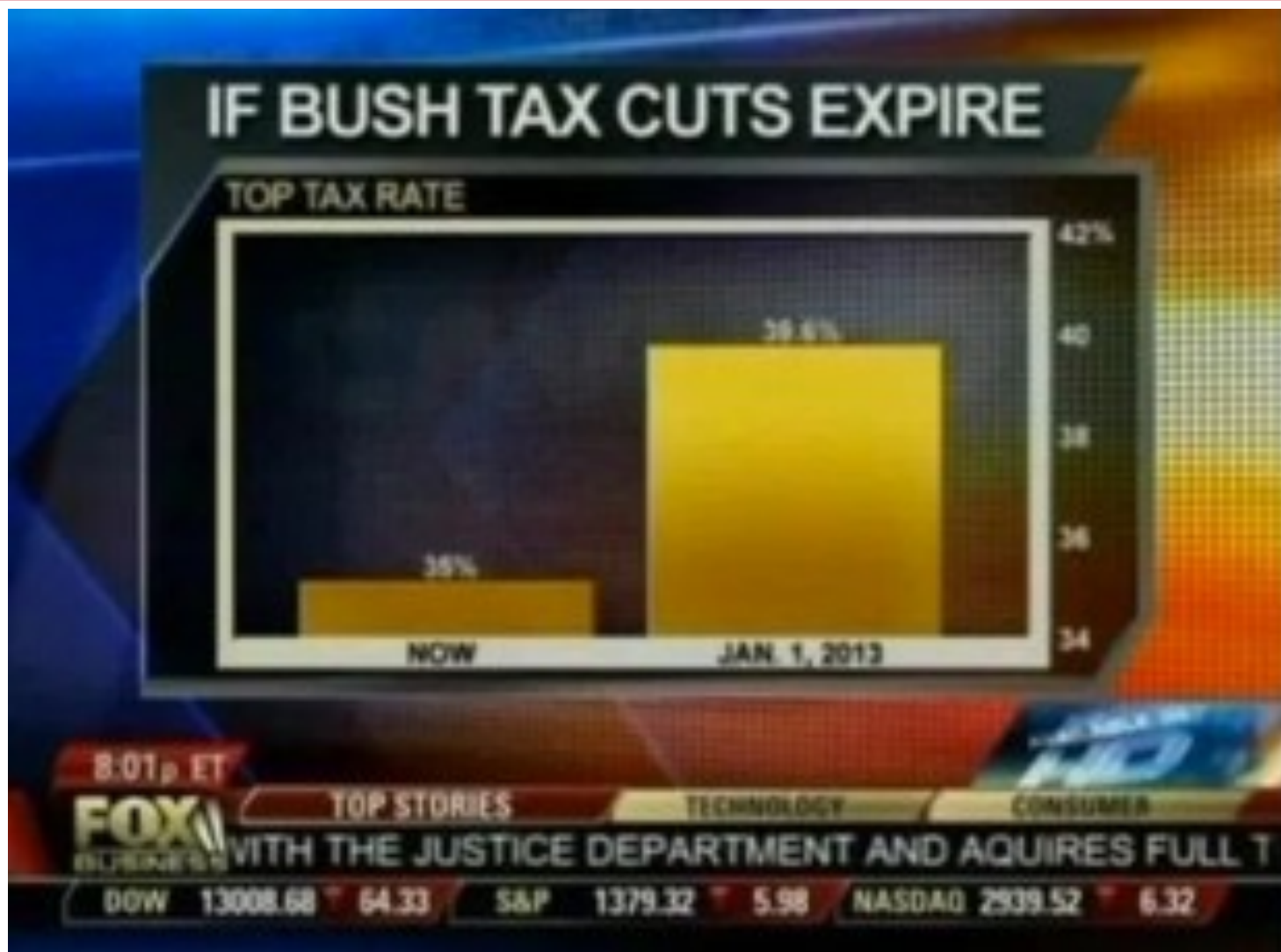


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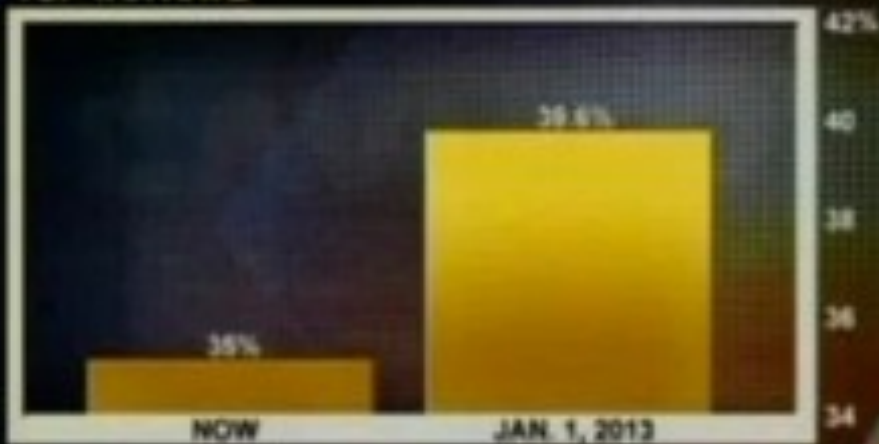
Edward Tufte's design principles

- Graphical integrity
- The lie factor
- Maximize data-ink ratio
- Avoid *harmful* chart junk



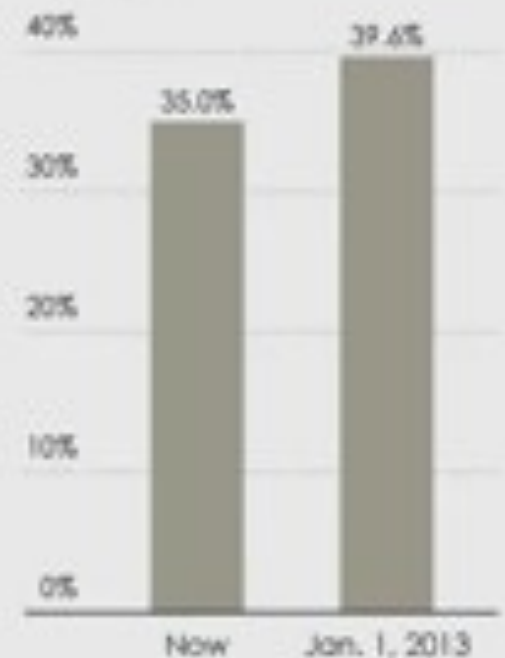
IF BUSH TAX CUTS EXPIRE

TOP TAX RATE

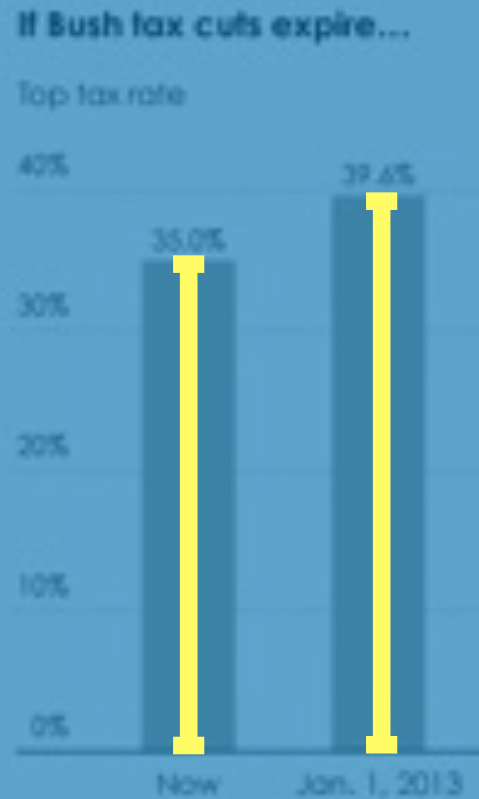


If Bush tax cuts expire...

Top tax rate



Bar Chart should have a **zero-baseline**.



Because you are comparing the **lengths**

Flowing data



Or comparing **positions** from the **baseline**.

Flowing data

Do we always need a zero-baseline?



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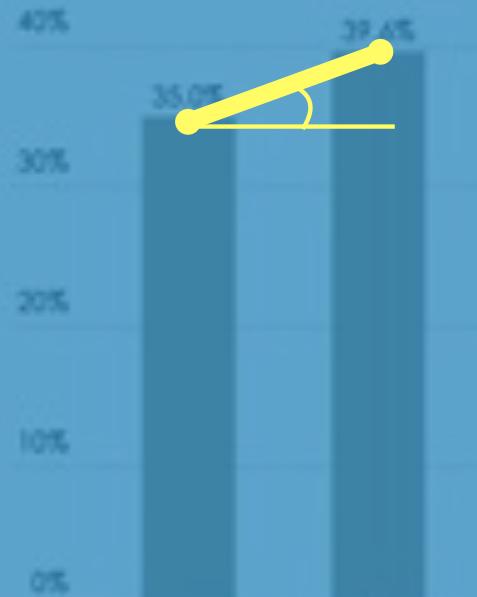
"USE A BASELINE THAT SHOWS THE DATA, NOT THE ZERO POINT."

Edward Tufte

Line Chart may **not** have a **zero-baseline**.

If Bush tax cuts expire...

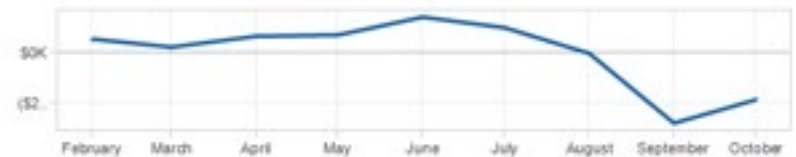
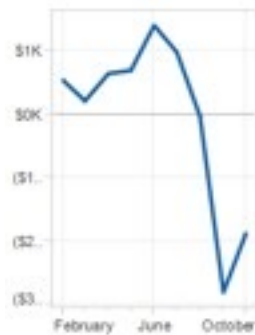
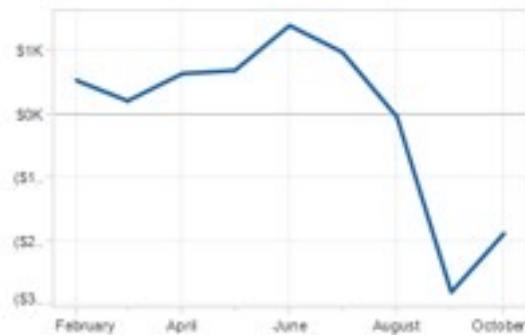
Top tax rate



Because you are comparing the **angle**

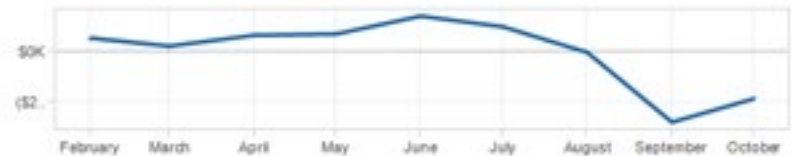
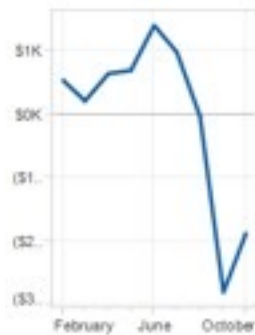
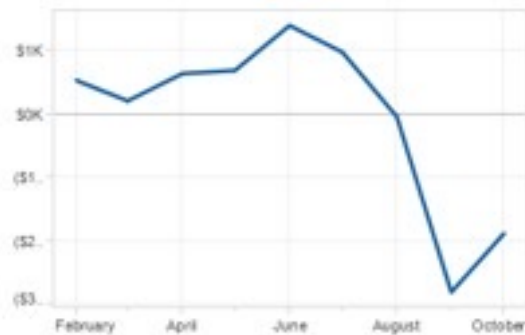
Flowing data

Line chart's aspect ratios can matter too



<https://eagereyes.org/basics/banking-45-degrees>

Line chart's aspect ratios can matter too



A rule of thumb is banking to 45 degrees (the average line slope in a line chart should be 45) to minimize errors in visual judgments of slope ratios.

JOB LOSS BY QUARTER



FOX NEWS
FOX NEWS
.COM

SOURCE: BLS

AMERICA'S
NEWSROOM

JOB LOSS BY QUARTER



RANDOM QUARTER

FOX NEWS
FOX NEWS
.COM

SOURCE: BLS

AMERICA'S
NEWSROOM

**INCORRECT Y-
SCALE**



SOURCE: BLS

JOB LOSS BY QUARTER

MISLEADING TITLE



15 MIL JOB LOSS?

FOX NEWS
FOX NEWS
.COM

SOURCE: BLS

AMERICA'S
NEWSROOM

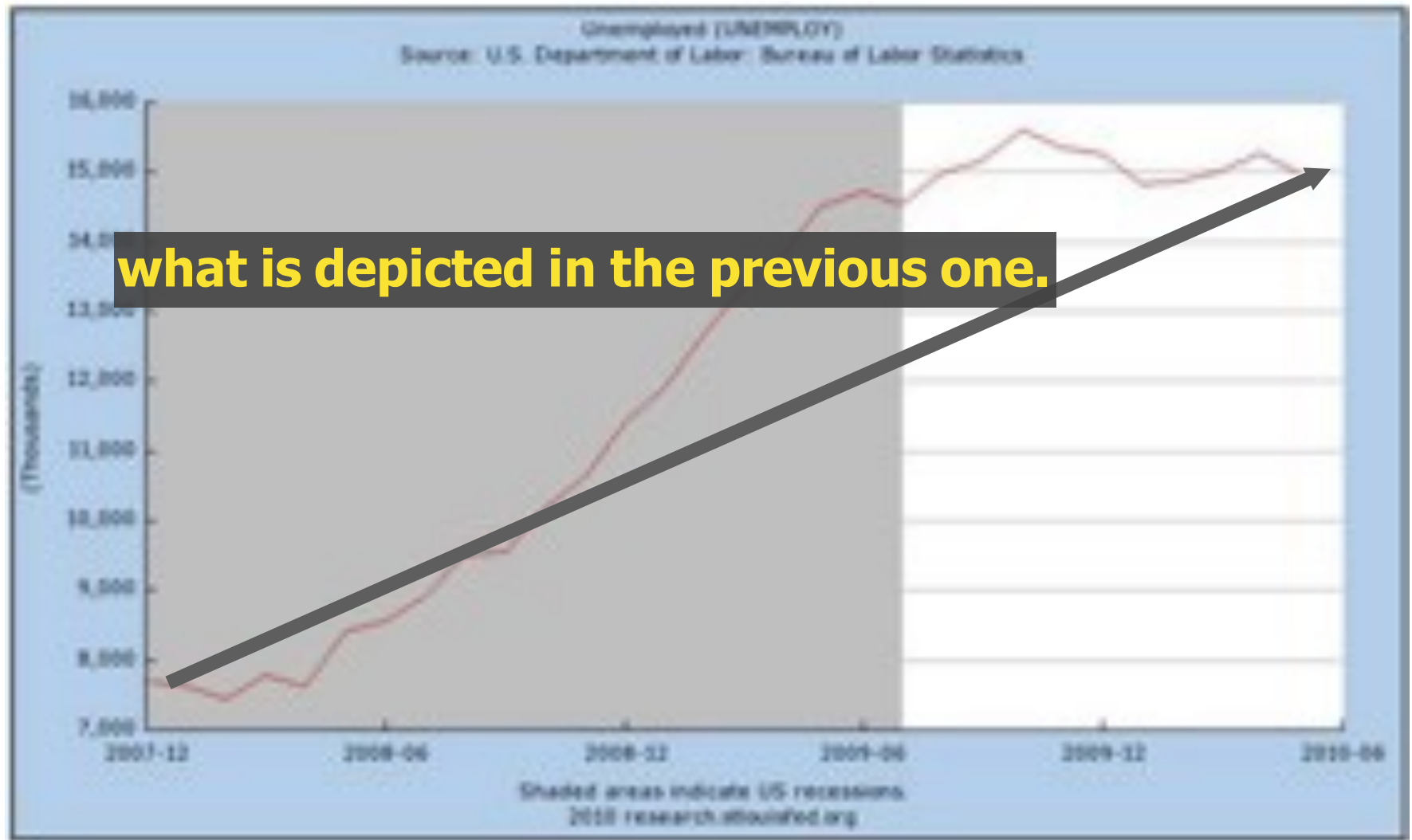
The truth
is...

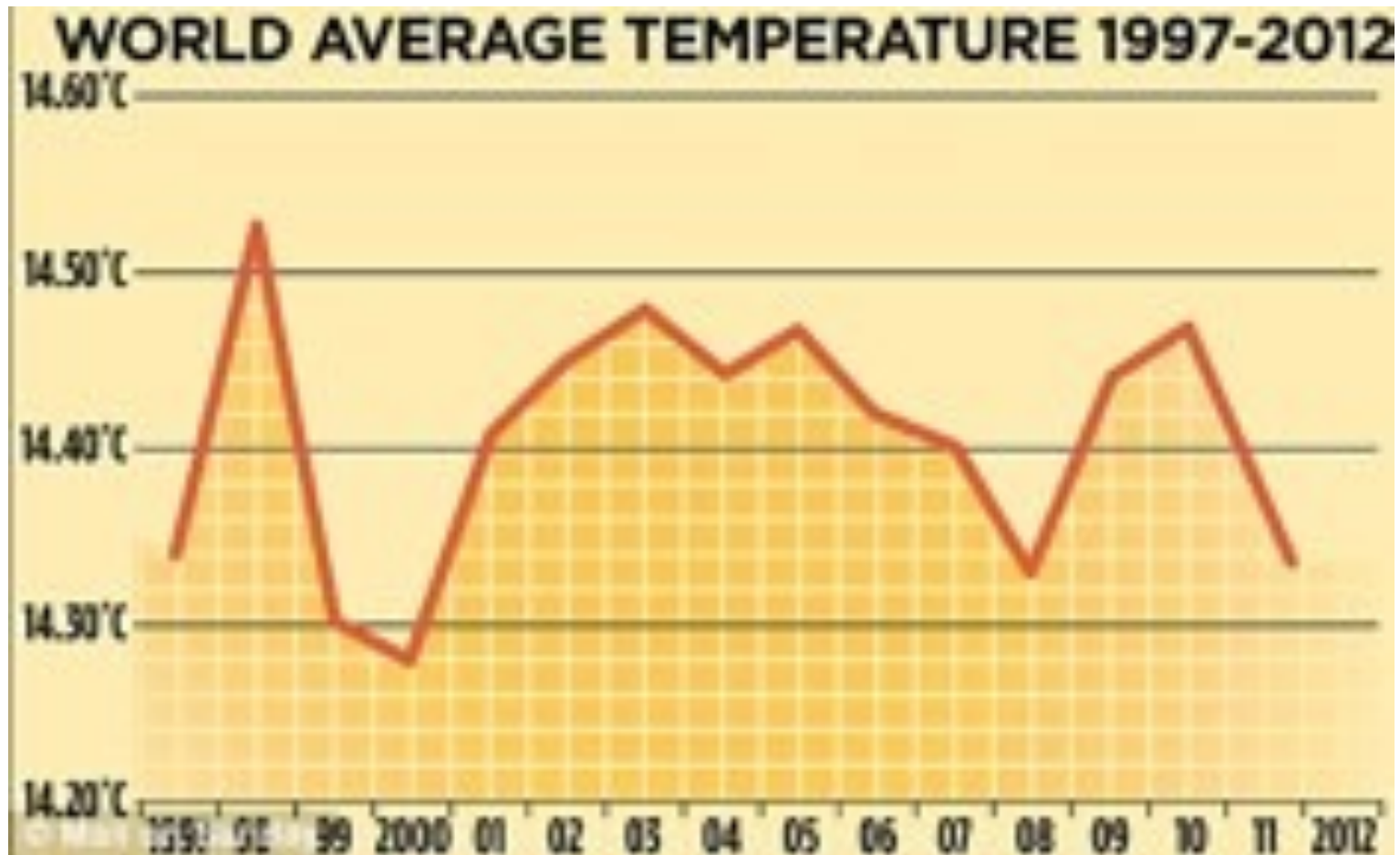
unemployed not job loss!

stabilized after 2009



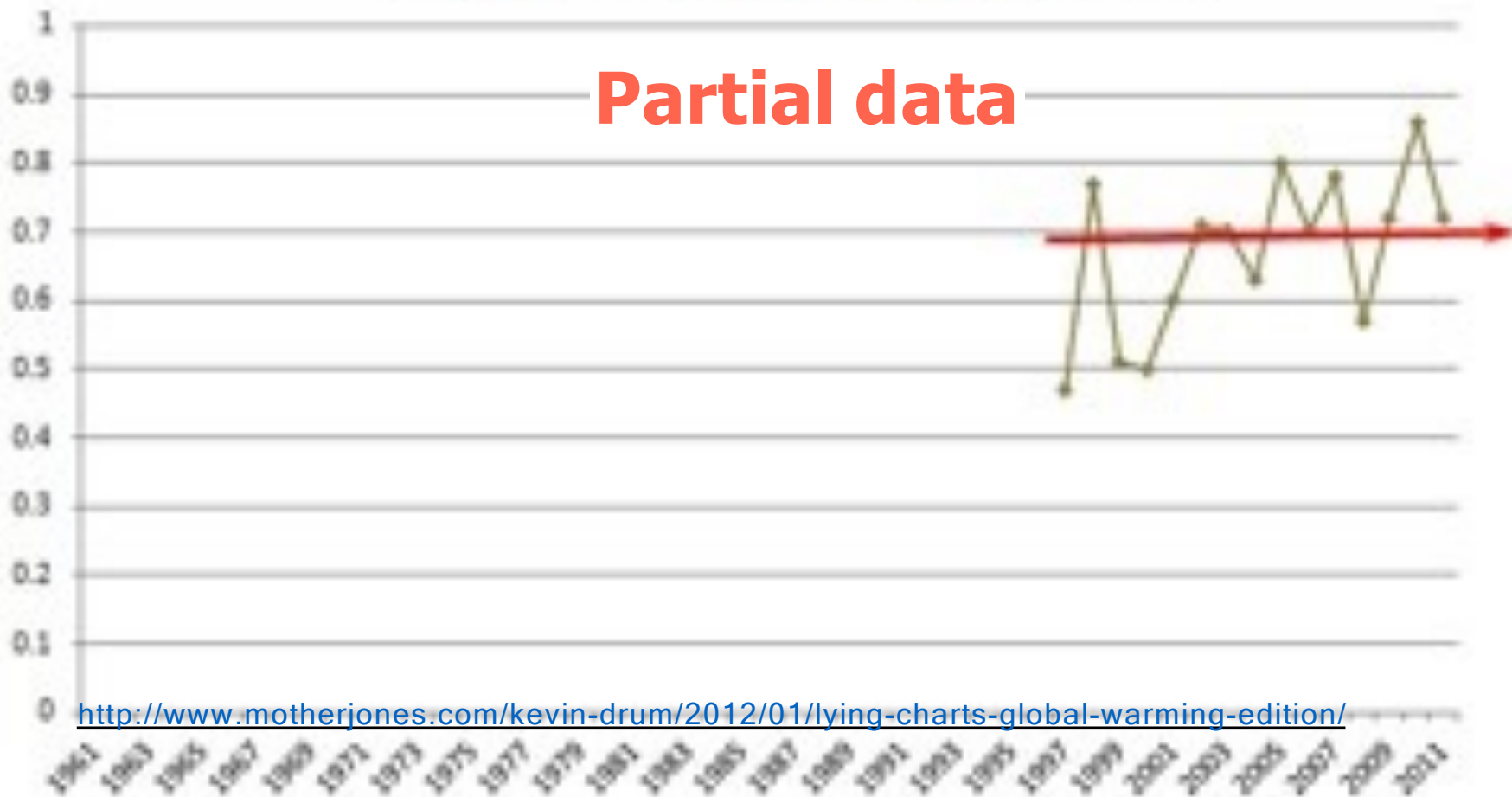
SOURCE: Worst chart I've seen all day, Jocelyn Fong (2010)





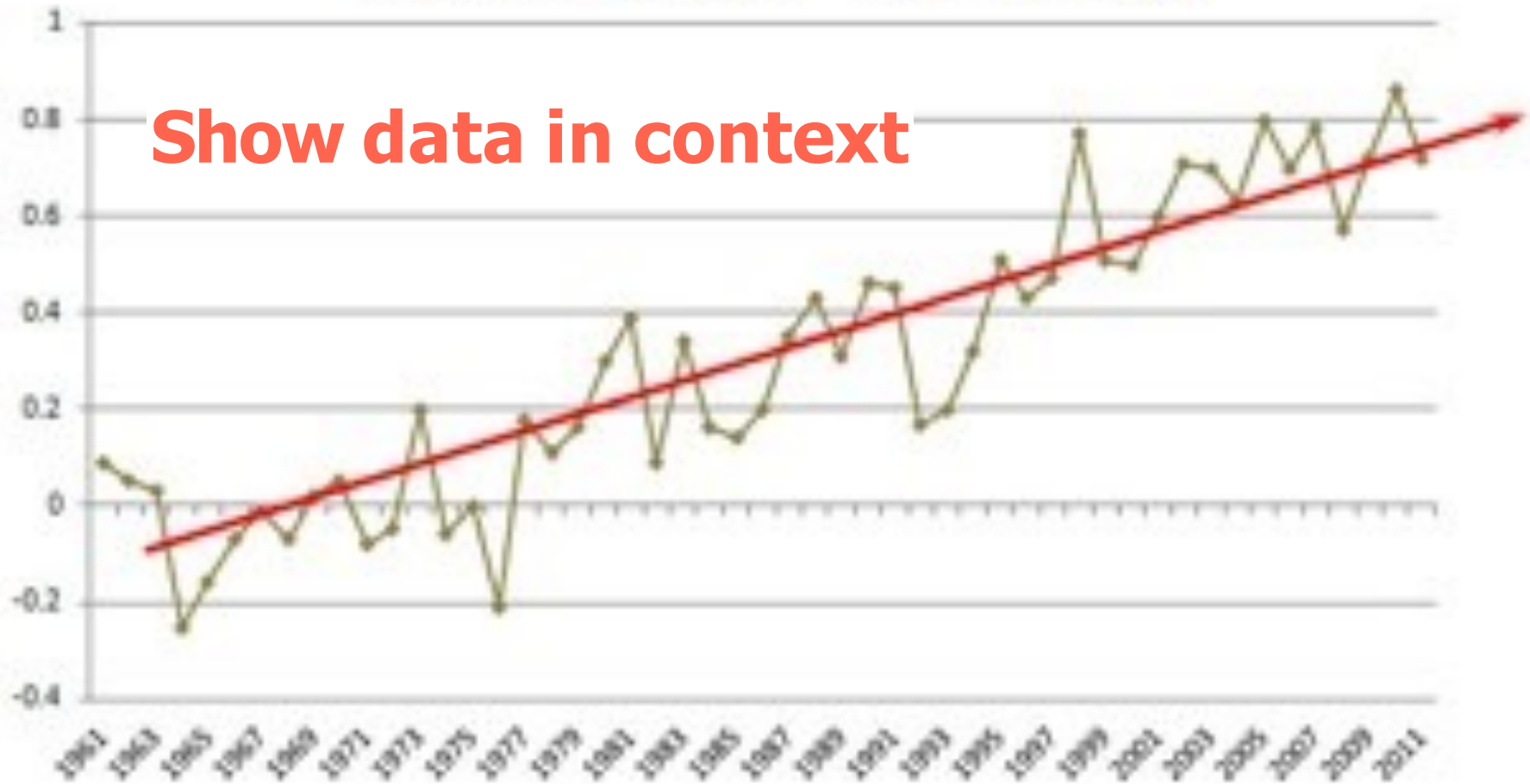
Temperature Anomaly -- Annual Mean (°C)

Partial data

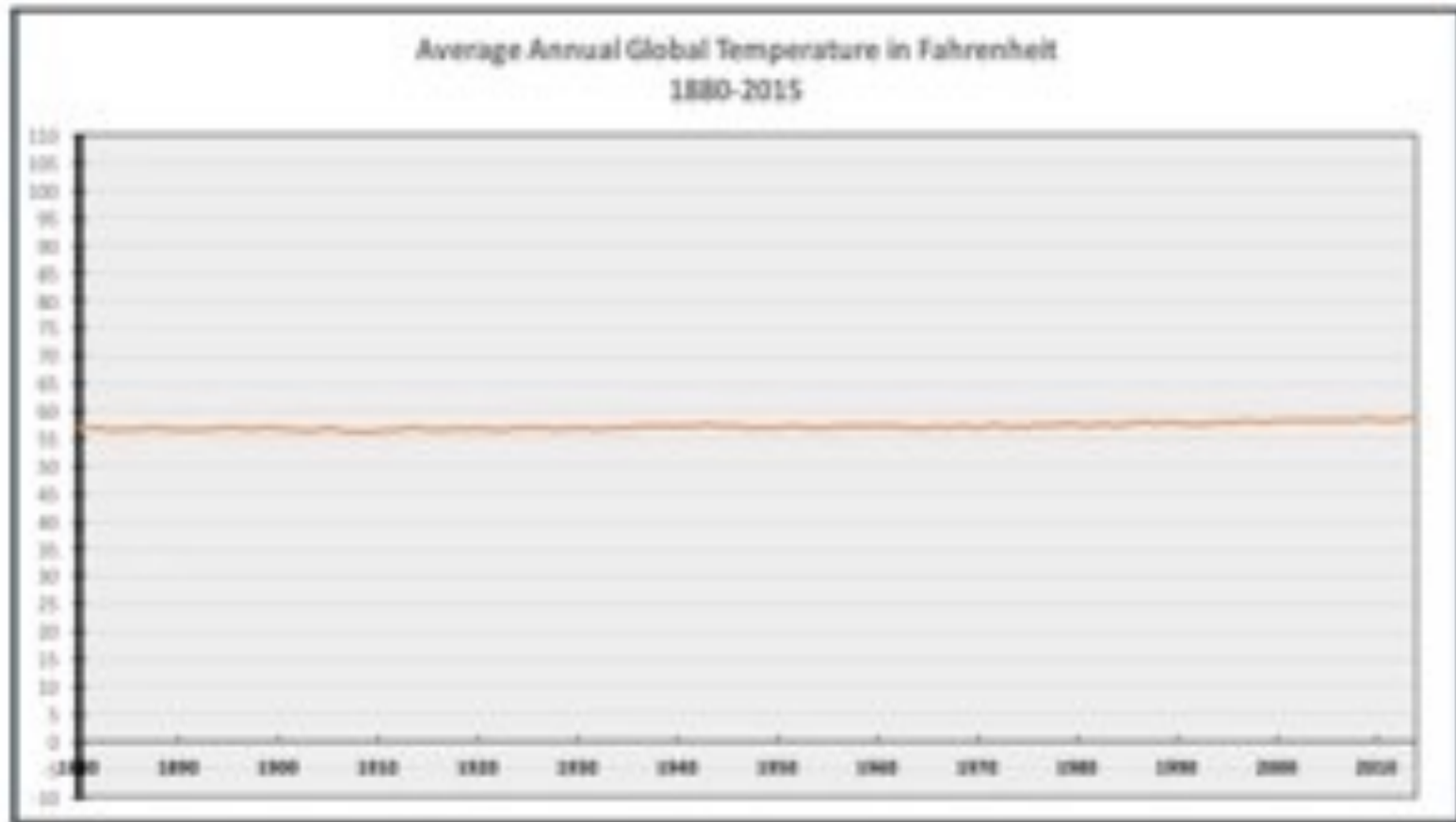


<http://www.motherjones.com/kevin-drum/2012/01/lying-charts-global-warming-edition/>

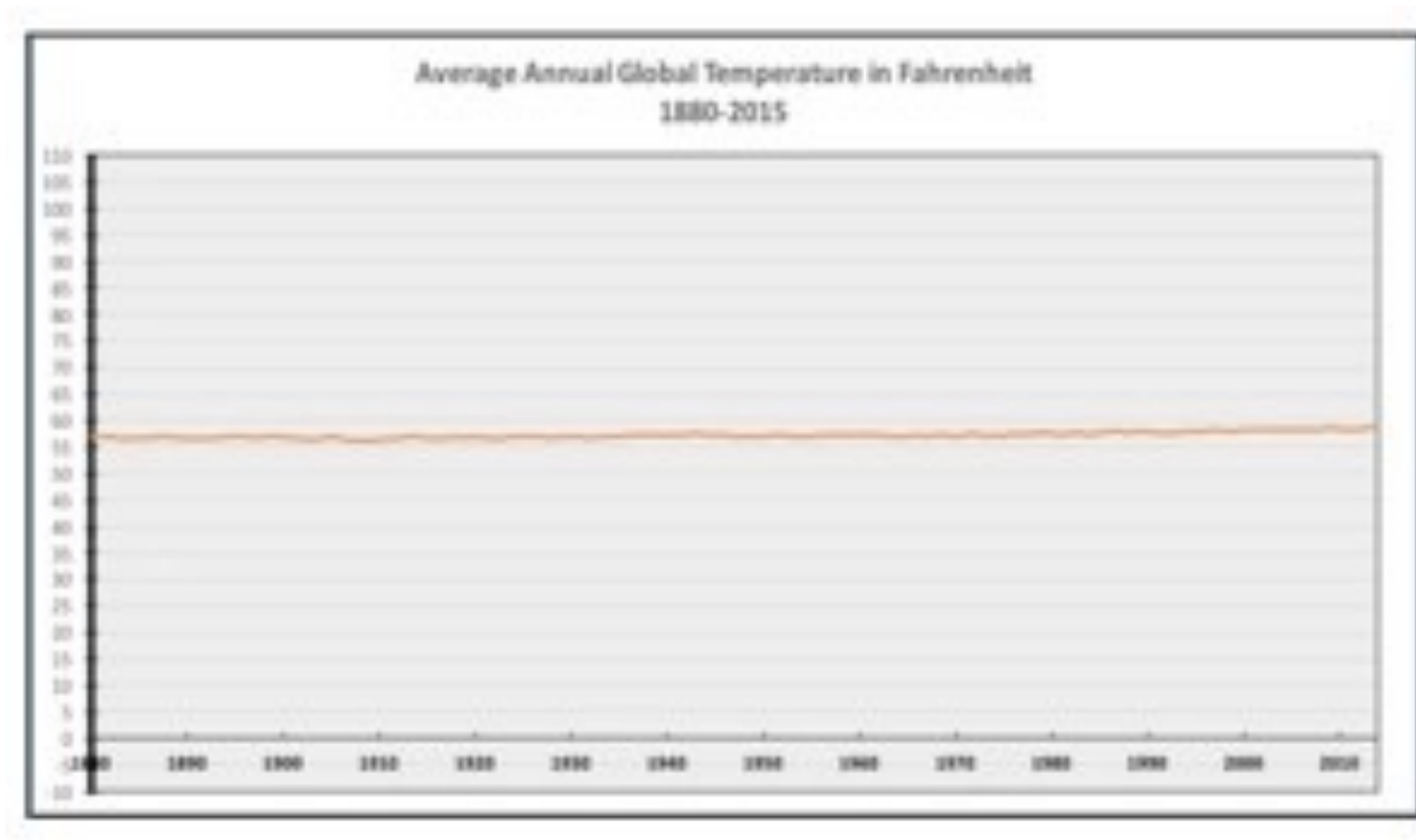
Temperature Anomaly -- Annual Mean (°C)



Average Annual Global Temperature (°F) 1880-2015



Choose axis scales wisely



Principles: Tell the truth

- The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the numerical quantities measured.
 - A classic case of misrepresentation of data is when a chart is created with a scale that is completely off.

MARKETING MACHINE

TOTAL JUVE
JERSEY'S SOLD
IN 2016

850K

RONALDO
JERSEYS SOLD
IN THE FIRST DAY

520K

NEYMAR
JERSEYS SOLD
FROM BARCA
TO PSG

10K



Lie factor

- The “Lie Factor” is a value to describe the relation between the size of effect shown in a graphic and the size of effect shown in the data.

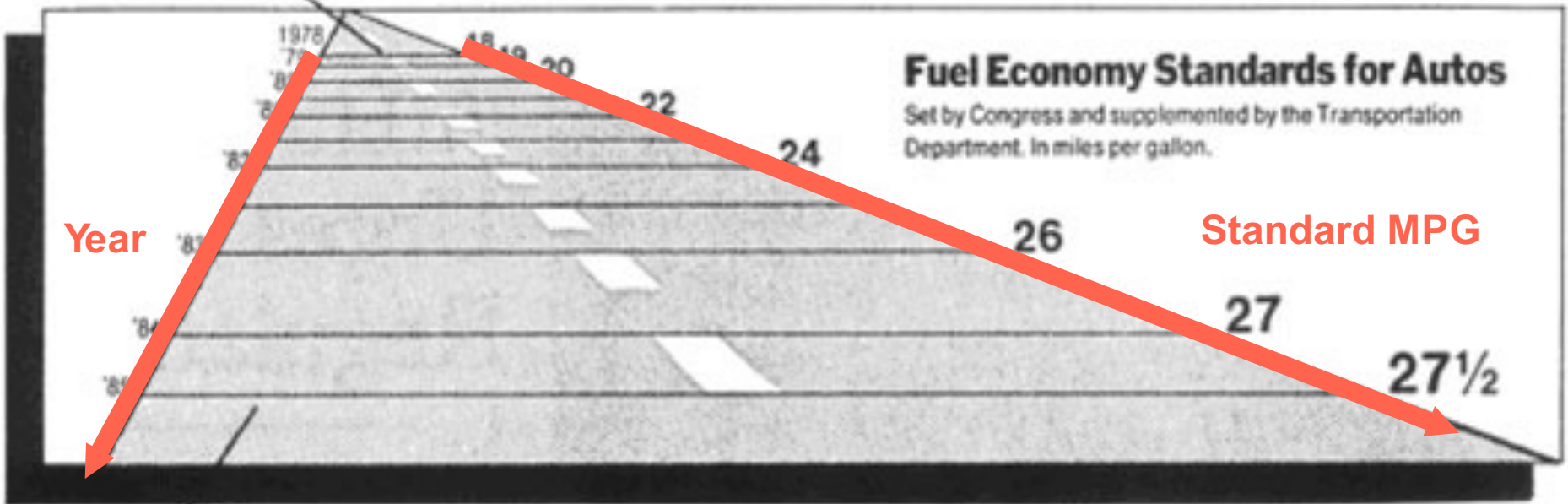
This results in the following formula:

$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$$

where

$$\text{size of effect} = \frac{|\text{second value} - \text{first value}|}{\text{first value}}$$

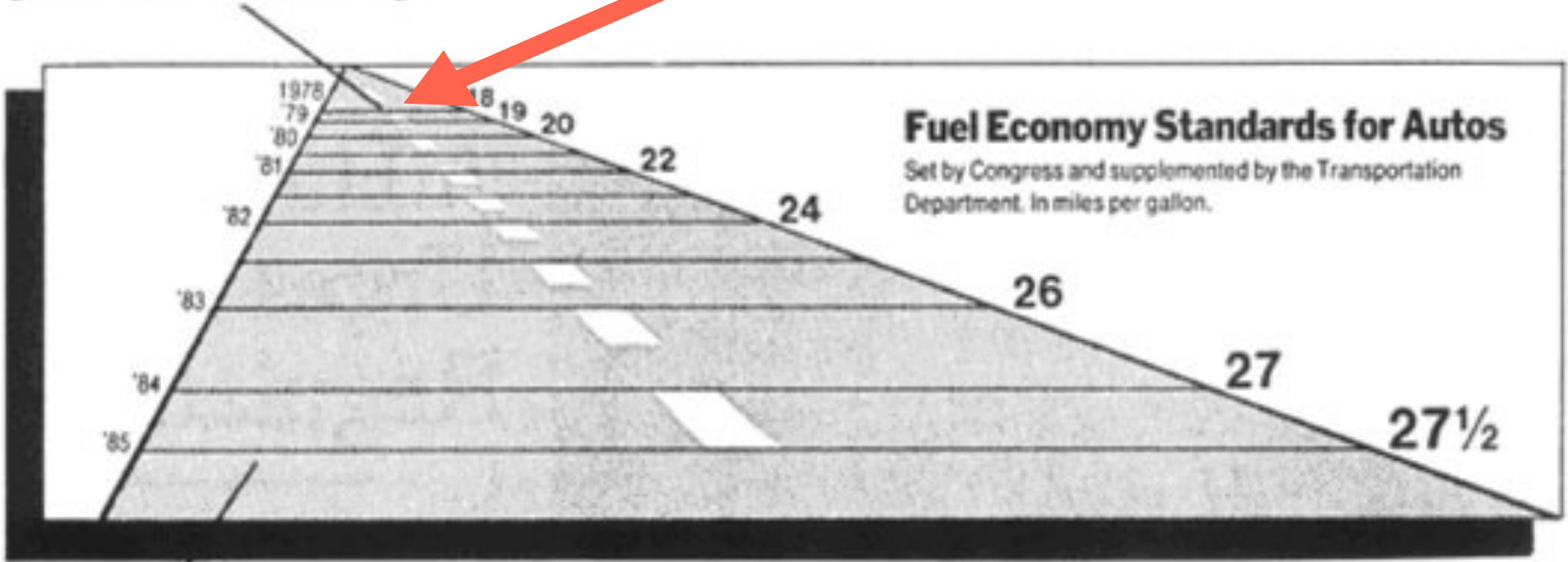
This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

This line, representing **18 miles** per gallon in 1978, is **0.6 inches** long

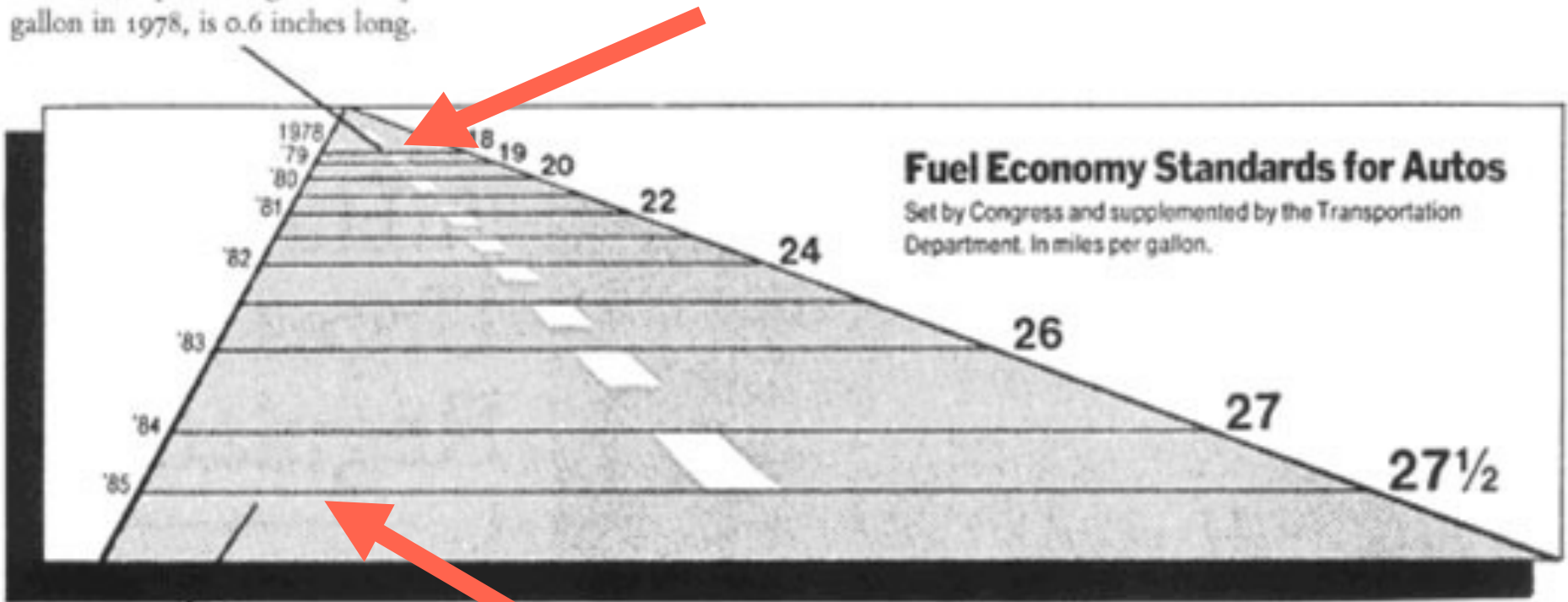
This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.

This line, representing **18 miles** per gallon in 1978, is **0.6 inches** long

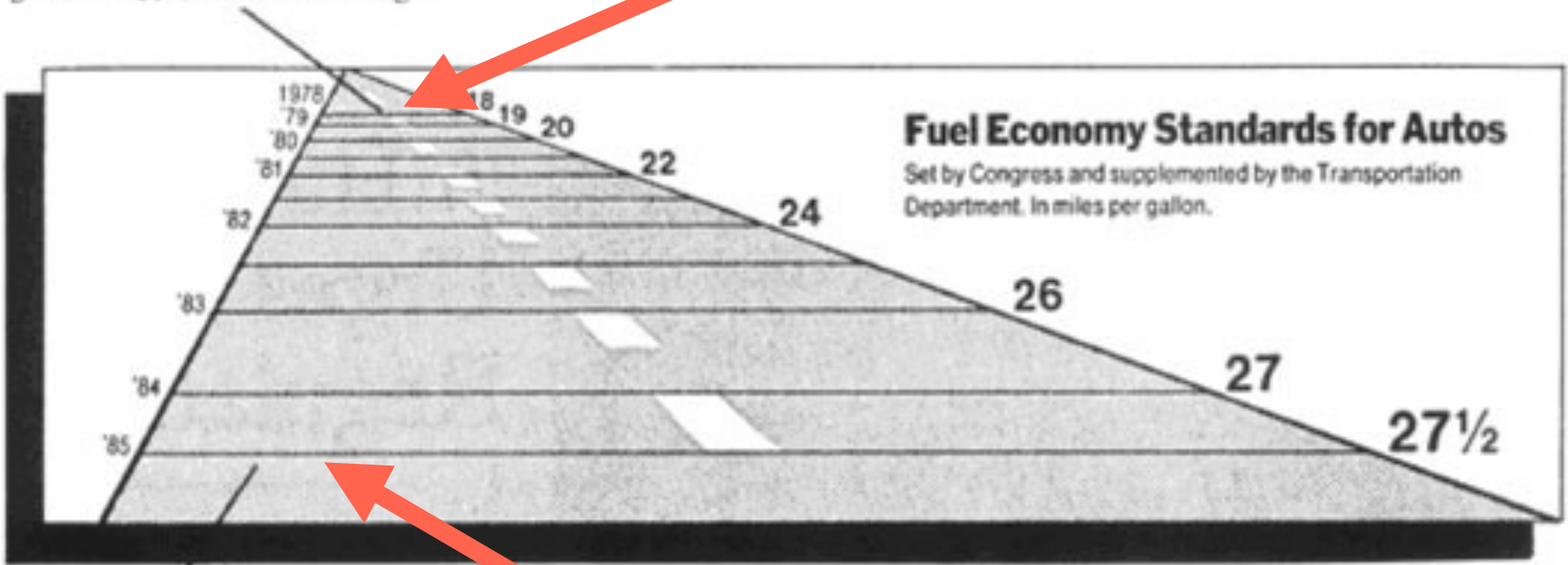


This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

This line, representing **27.5 miles** per gallon in 1985, is **5.3 inches** long

This line, representing **18 miles** per gallon in 1978, is **0.6 inches** long

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing **27.5 miles** per gallon in 1985, is **5.3 inches** long

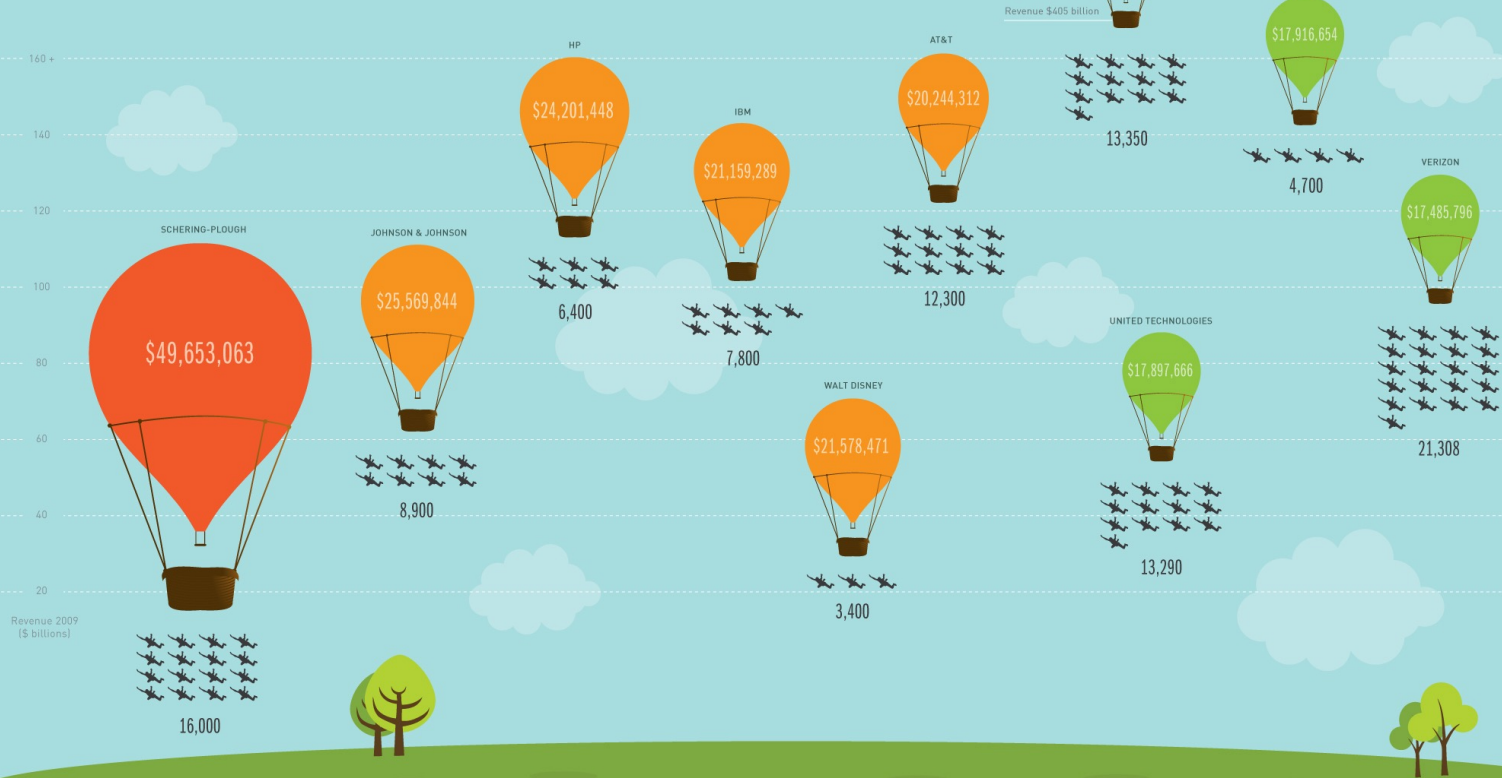
This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

$$\text{Lie Factor} = \frac{(5.3 - 0.6) / 0.6}{(27.5 - 18) / 18} = 14.8$$

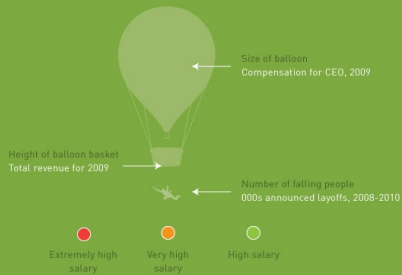
BALLOONING CEO SALARIES AND MASS LAYOFFS

New research suggests that CEOs are still making millions, despite cutting thousands of employees due to cost savings. According to the 17th annual Executive Excess report by the Institute for Policy Studies, the CEOs of the 50 US firms that cut the most jobs between November 2009 and April 2010 took 42 percent more than the average CEO at an S&P 500 firm. Here we take a look at the CEO compensation of the top 10 great recession layoff leaders...

Meet The Boss™



KEY



SLASHING JOBS PAYS



CEO SALARIES OVER TIME

Average annual CEO salary (adjusted to 2000 dollars)



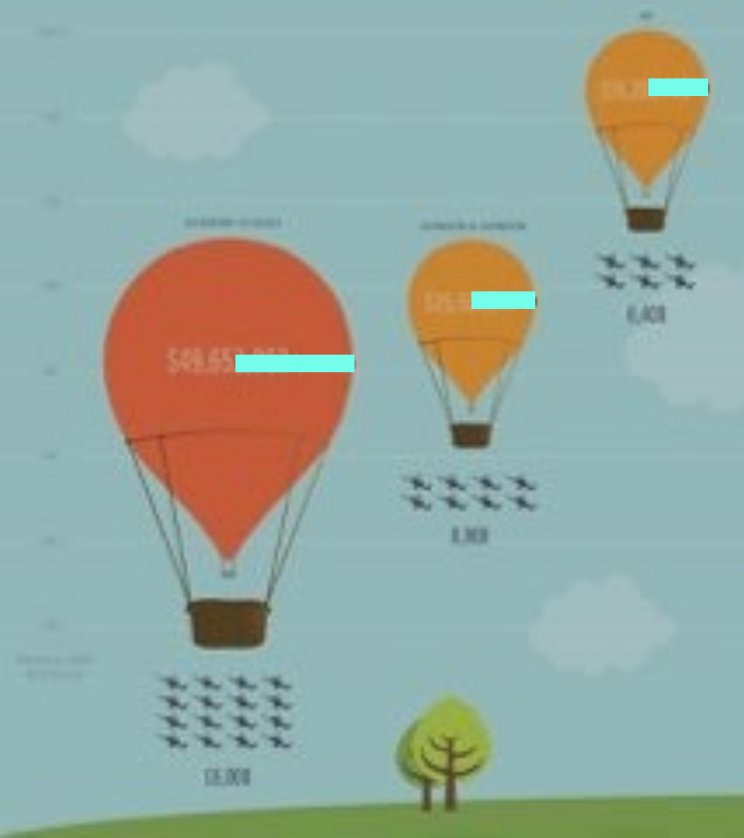
CEO PAY AND UNEMPLOYMENT INSURANCE

Combined compensation of top 50 CEOs in the layoff leader survey



BALLOONING CEO SALARIES AND MASS LAYOFFS

THE LARGEST COMPANY IN THE
WORLD, GOOGLE, HAS A CEO
SALARY OF \$49.65 MILLION
PER YEAR. THAT'S 10 TIMES
THE SALARY OF THE
SMALLEST COMPANY, MICRO
SOFT, WHICH HAS A CEO
SALARY OF \$4.965 MILLION
PER YEAR.



This chart uses **radius** of the balloon to encode the data

Doubling the **radius** (or data) increases the **perceived area** by **four**.

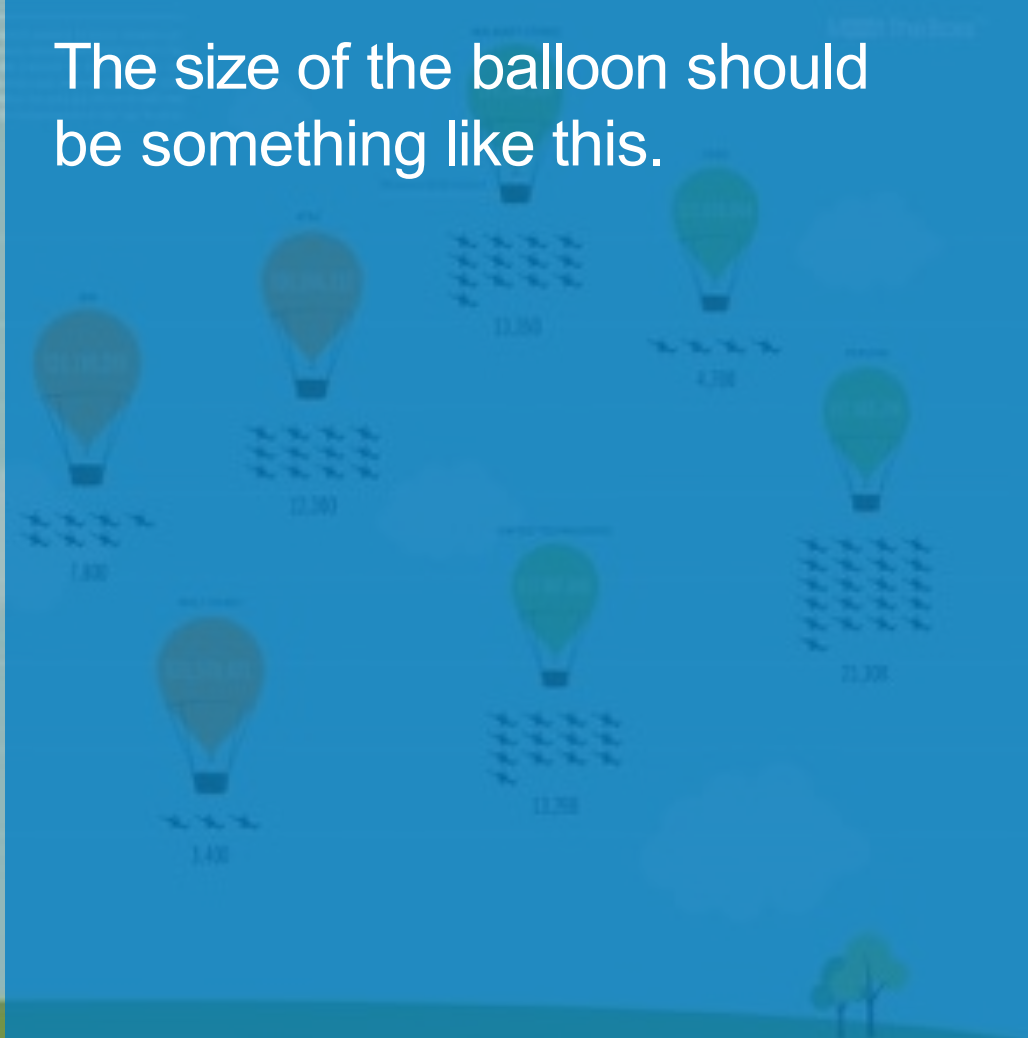
Lie factor = $(4-1)/(2-1) = 3$.

BALLOONING CEO SALARIES AND MASS LAYOFFS

THE LARGEST COMPANIES IN THE
WORLD HAVE A TENDENCY TO
PAY CEOs SALARIES THAT ARE
HUGELY OUT OF LINE WITH THE
SALARIES OF THE OTHER
EMPLOYEES.



The size of the balloon should be something like this.

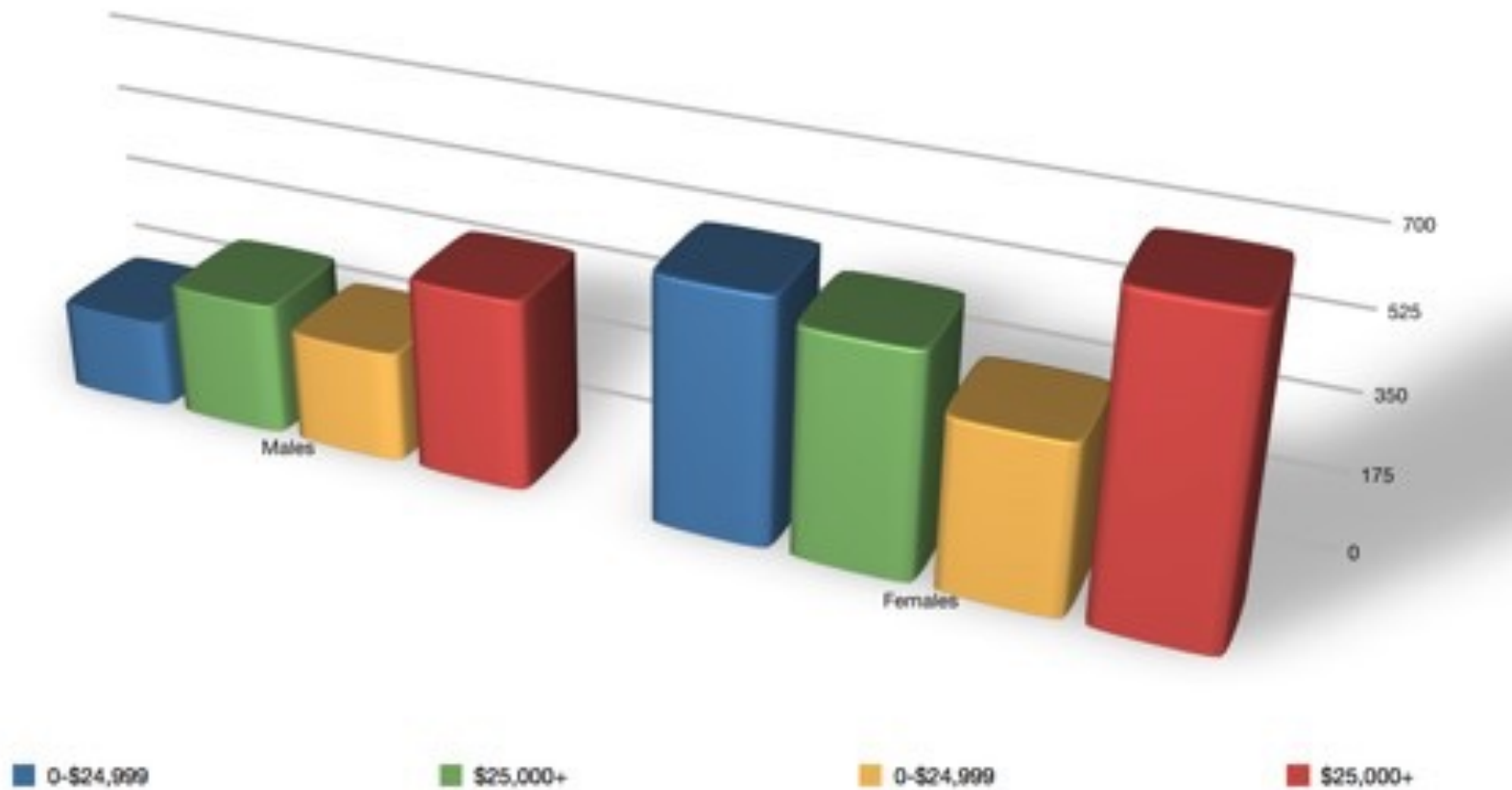


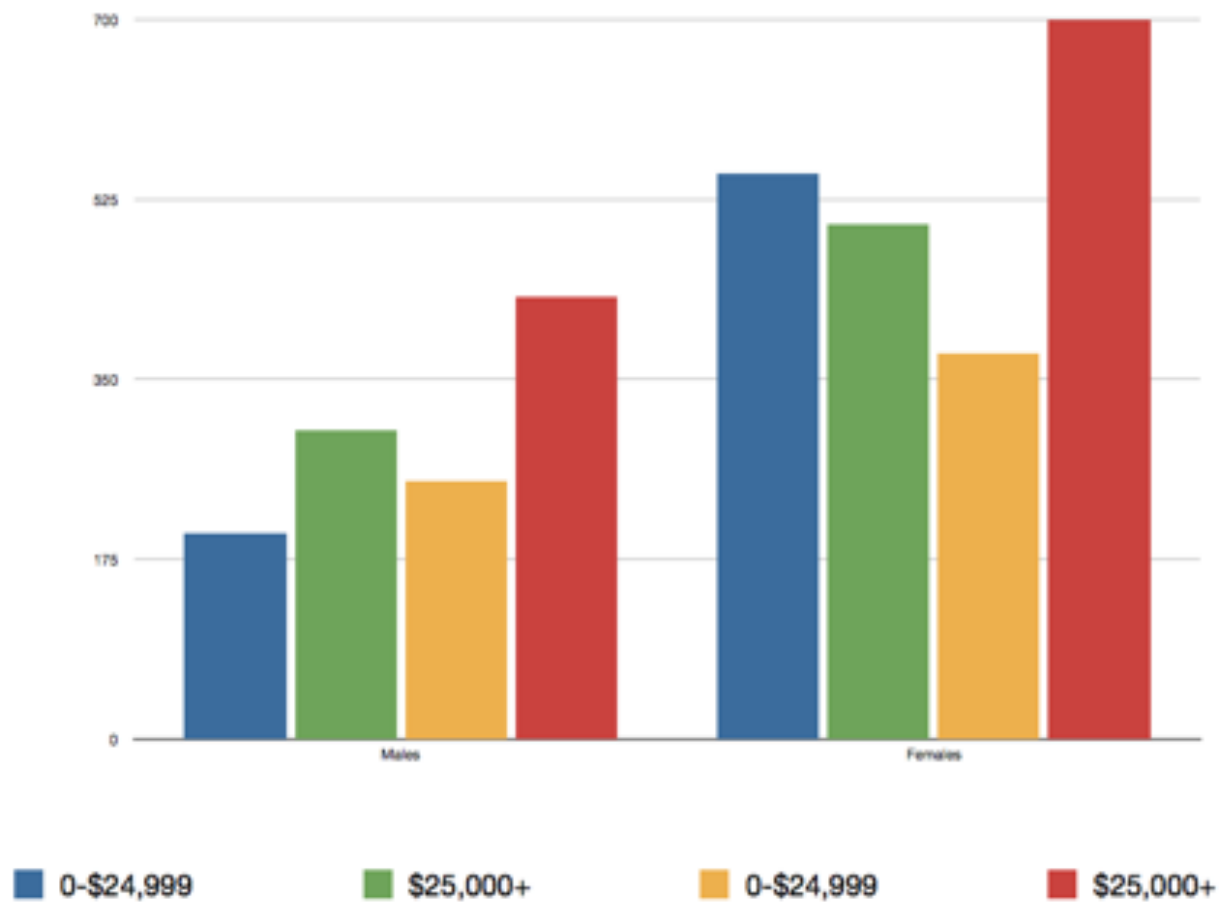
Principles: Maximize data-ink ratio

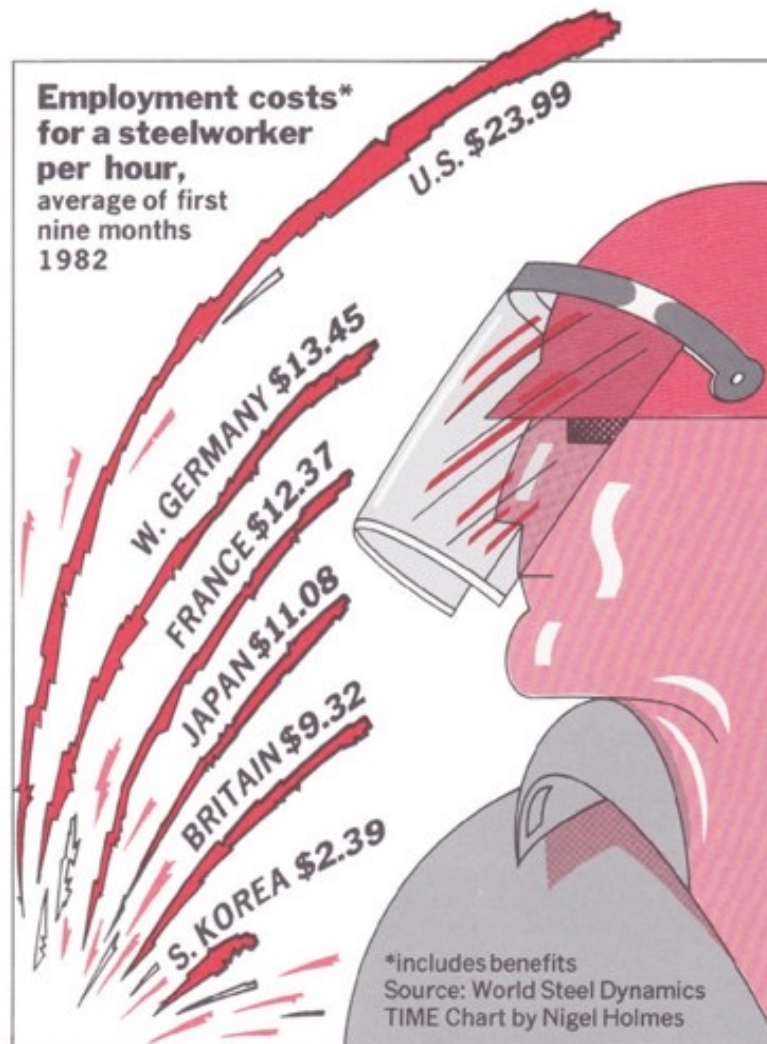
- Use the least ink to represent the greatest amount of information in the smallest space

Maximize Data-Ink Ratio

$$\begin{aligned}\text{Data-Ink Ratio} &= \frac{\text{Data-Ink}}{\text{Total "ink" used to print the graphic}} \\ &= \text{Proportion of a graphic's "ink" devoted to the non-redundant display of data-information} \\ &= 1.0 - \text{proportion of a graphic that can be erased without loss of data-information}\end{aligned}$$

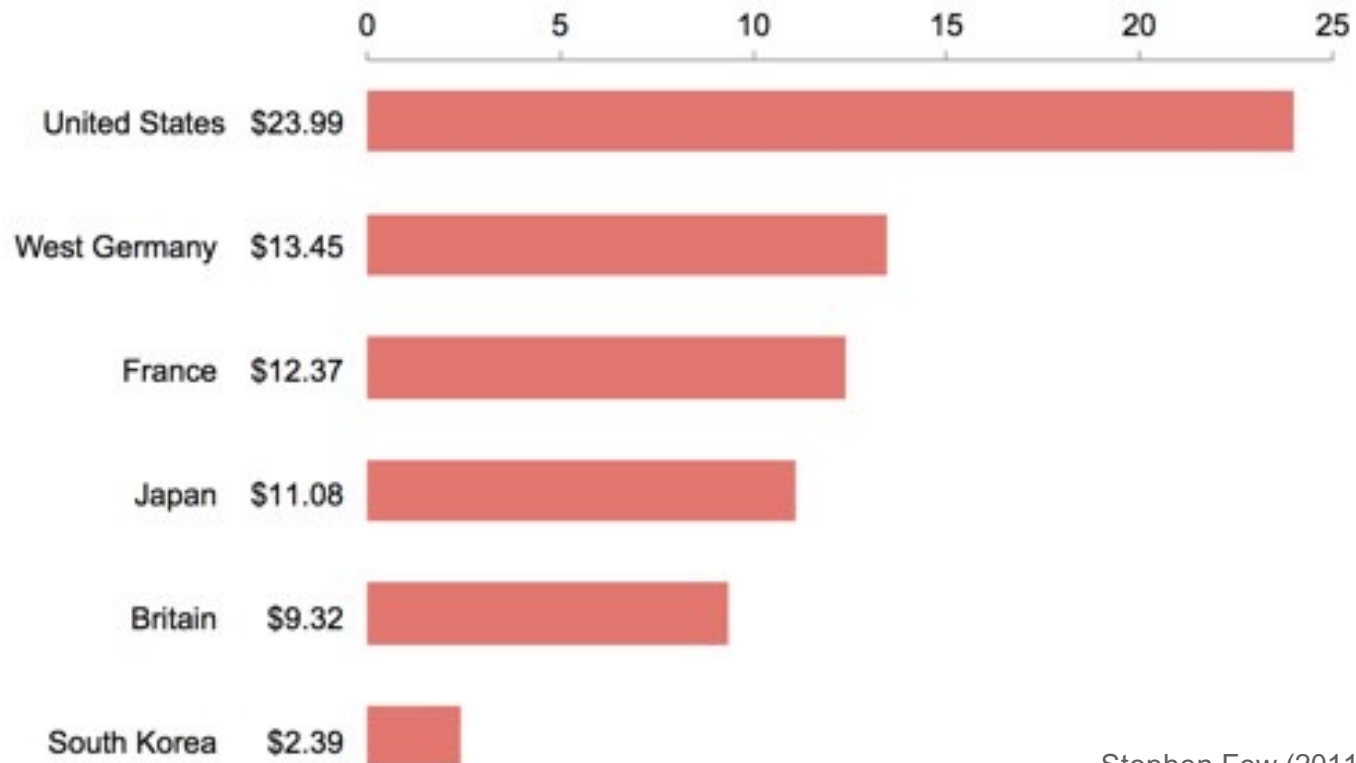






Employment Costs for a Steelworker per Hour

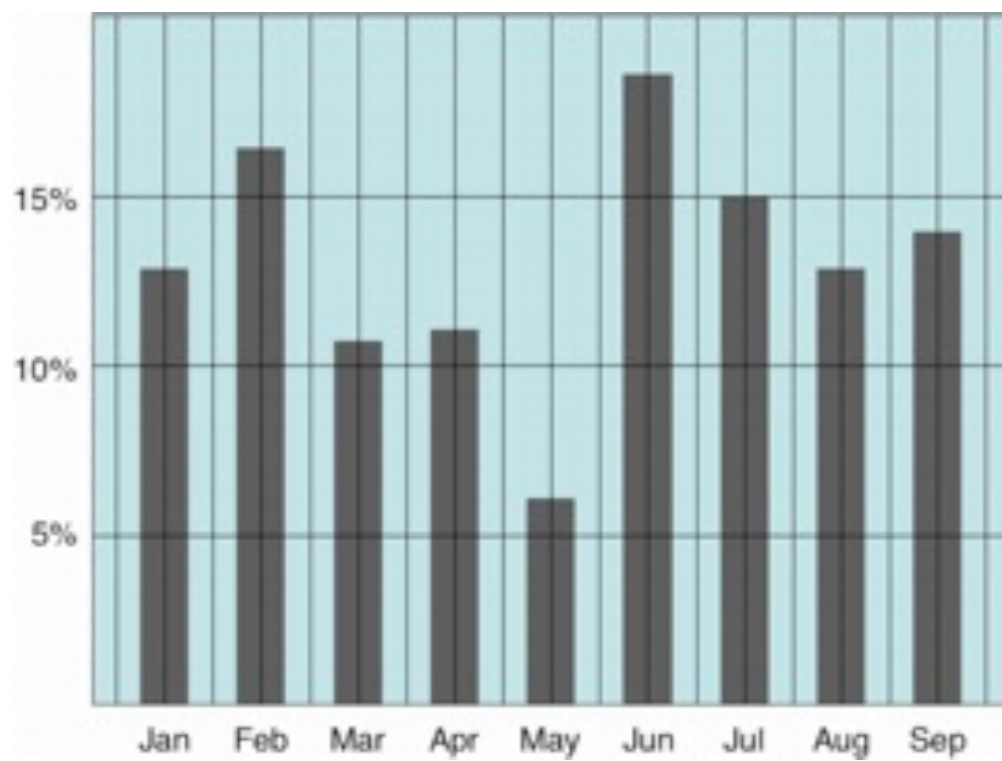
Average of first 9 months of 1982 in U.S. Dollars

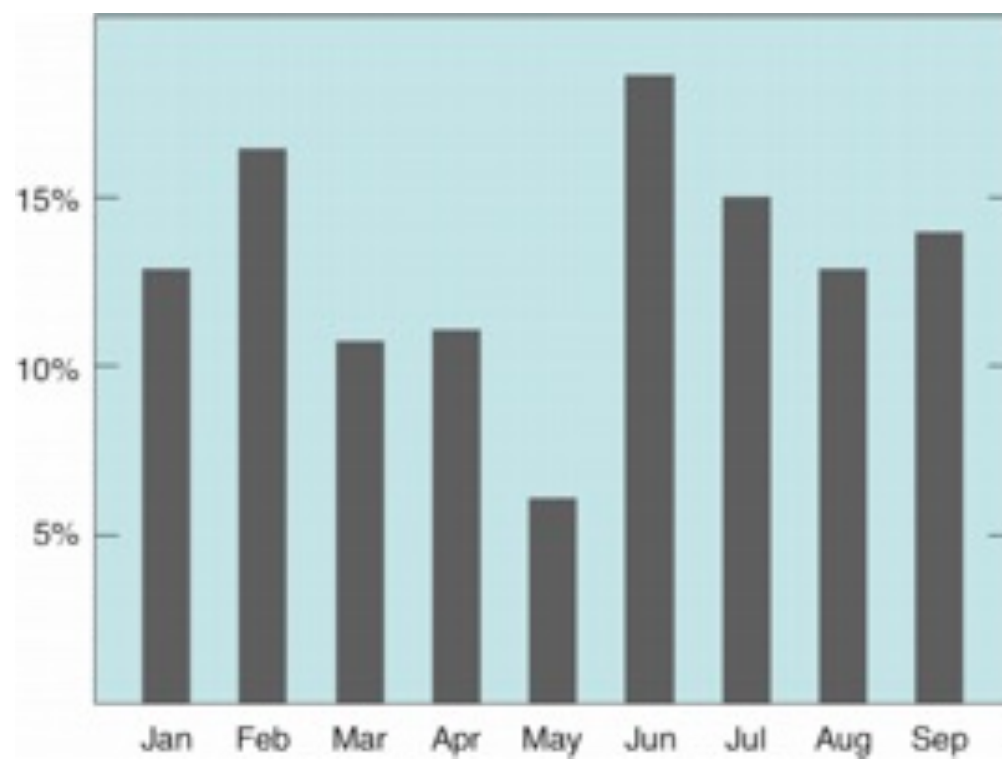


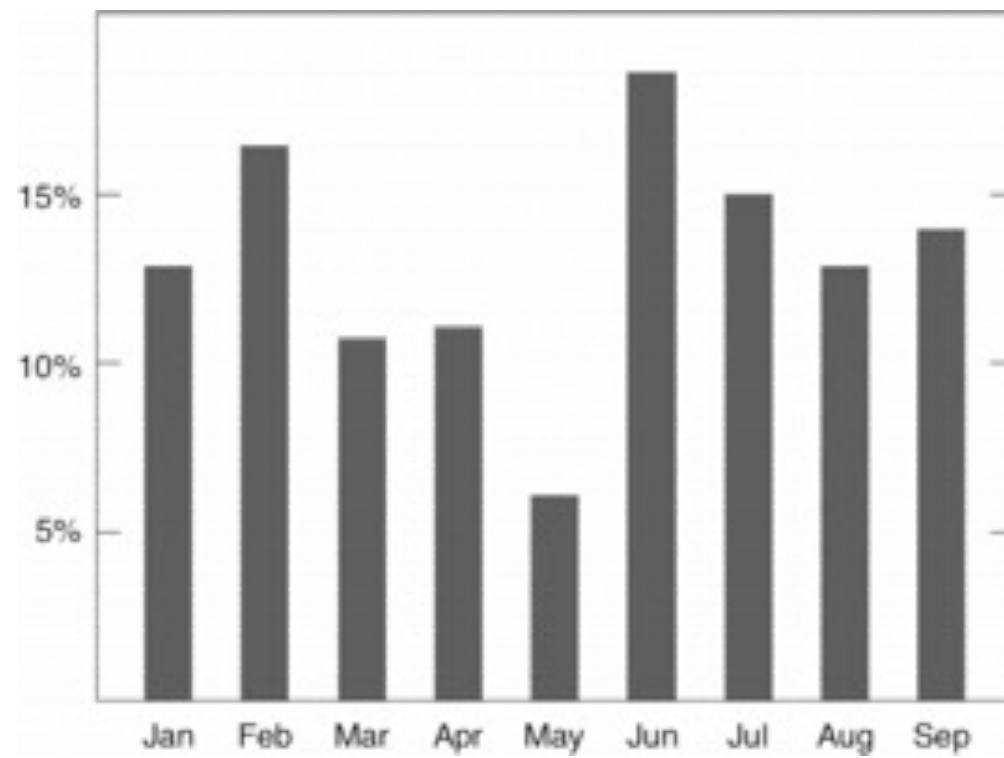
Stephen Few (2011)

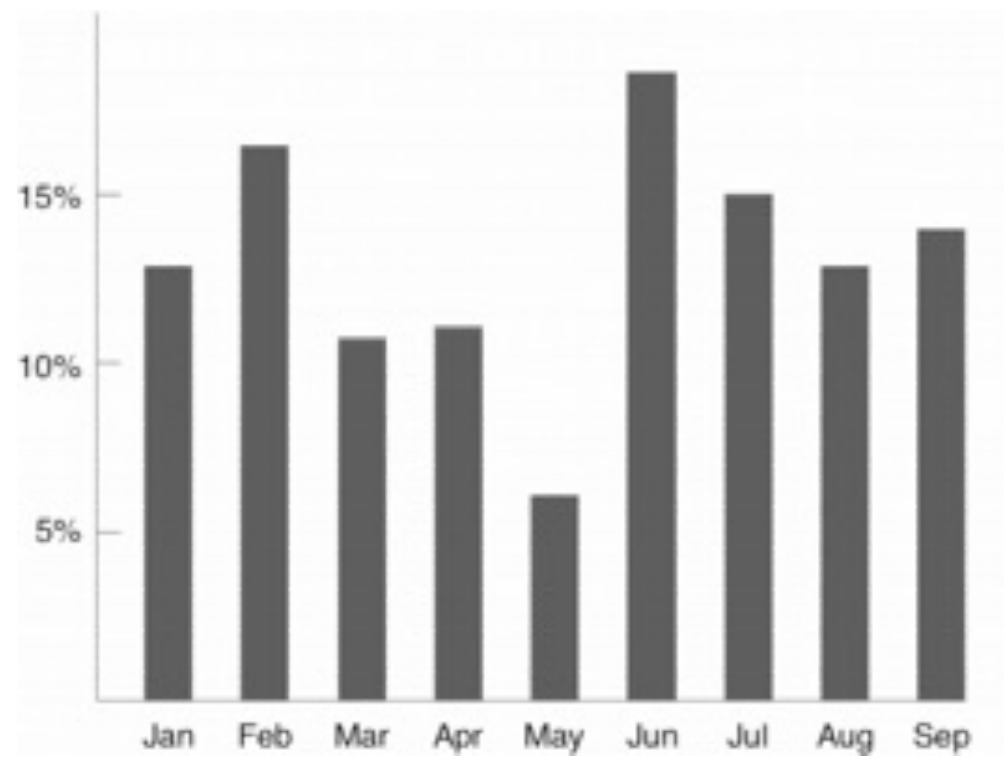
Principles: Avoid chart junks

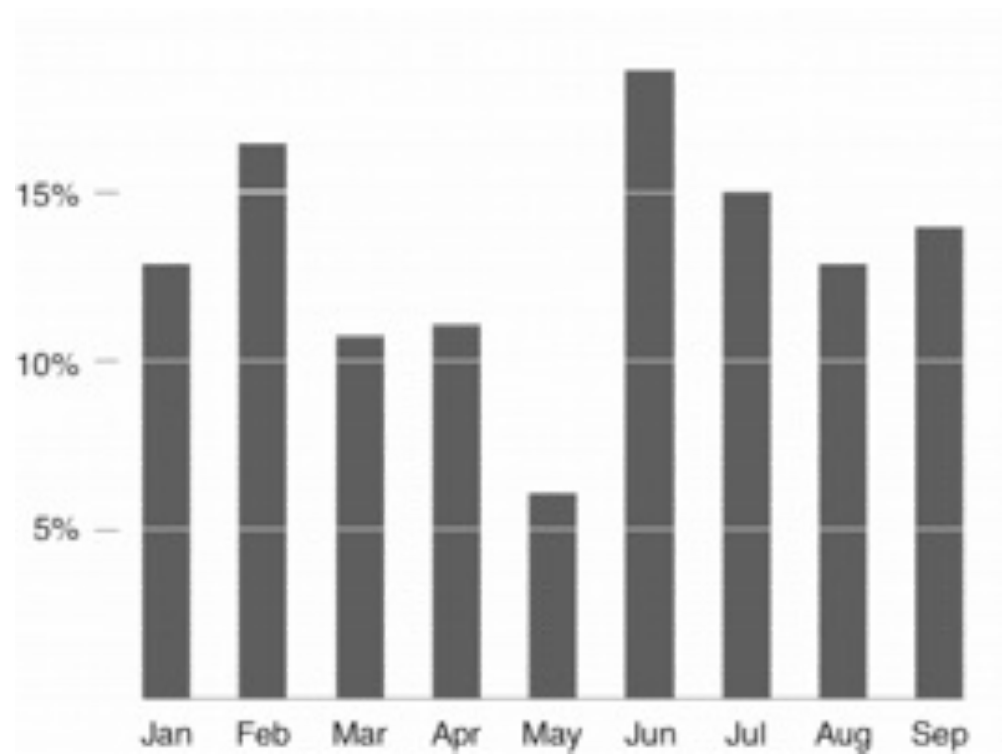
- Unnecessary visual elements in charts that distracts the viewer from the information.

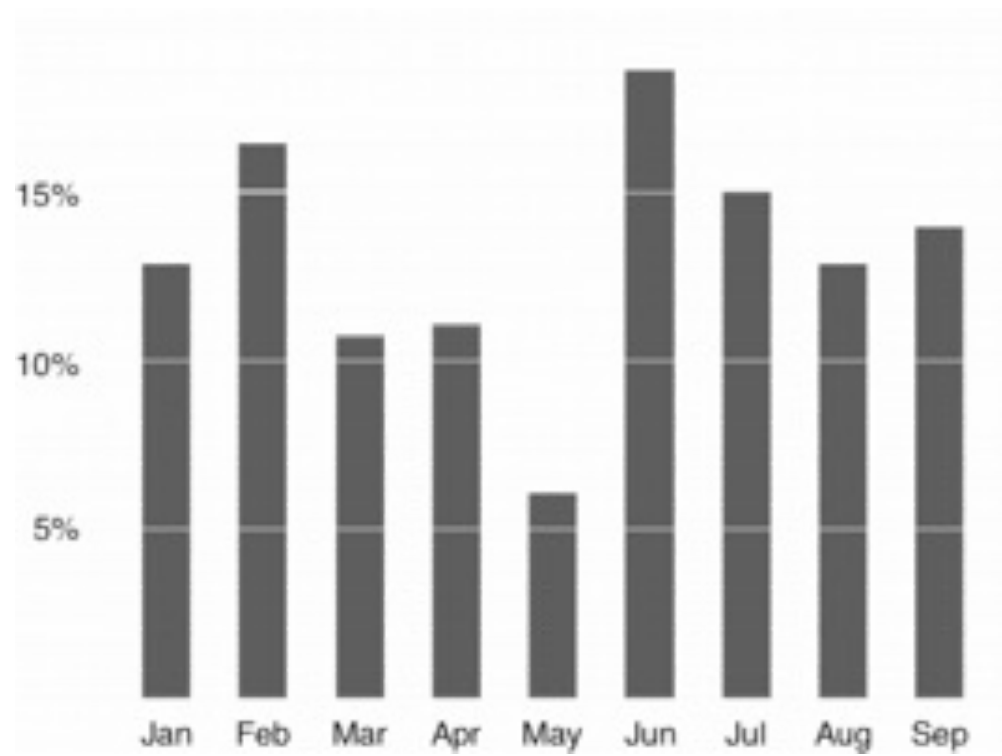




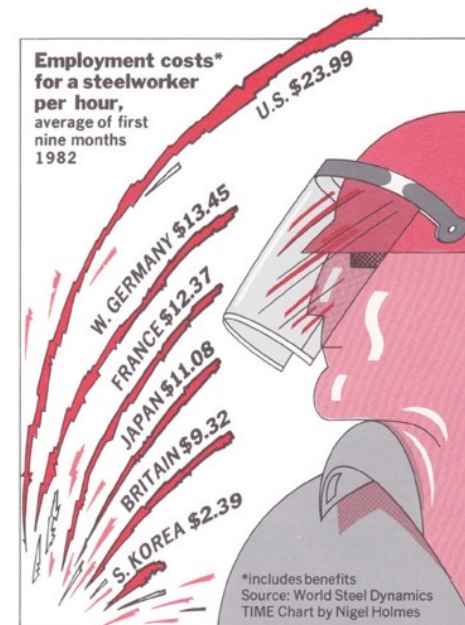








Are these chart junks?



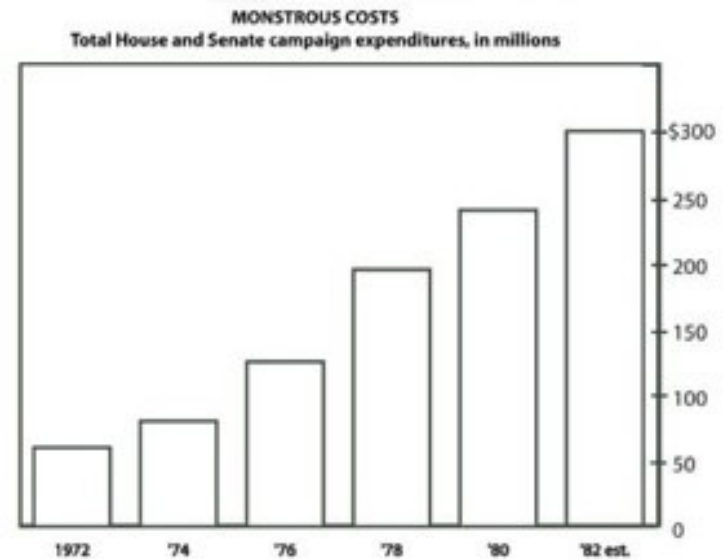
Not all chart junks are the same



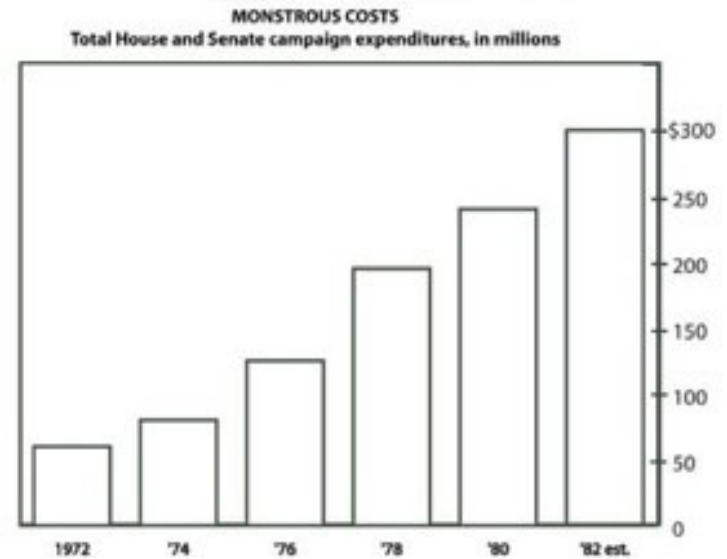
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Useful chart junks?



Not harmful in comprehension but more engaging & memorable



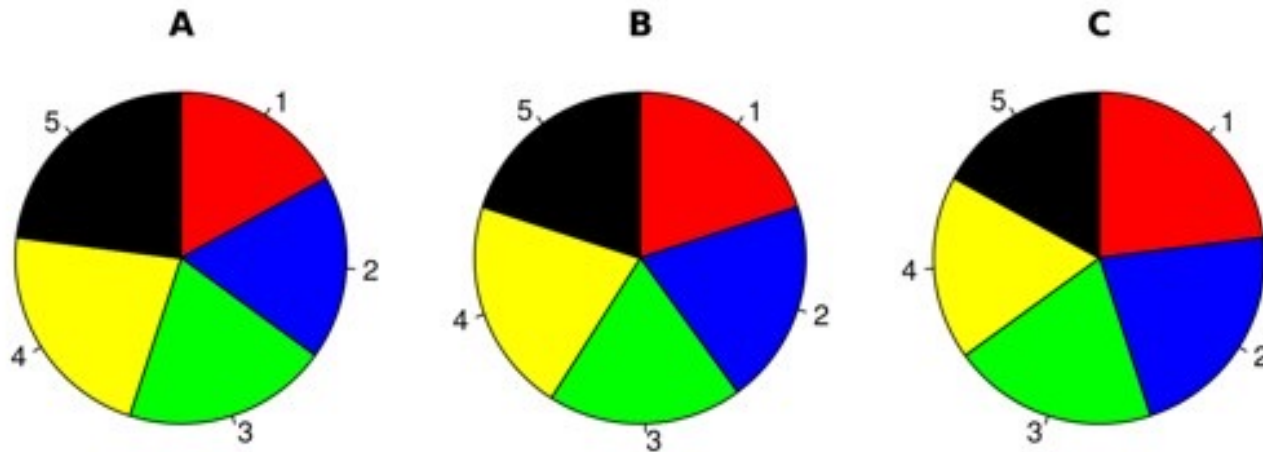
Pie Charts



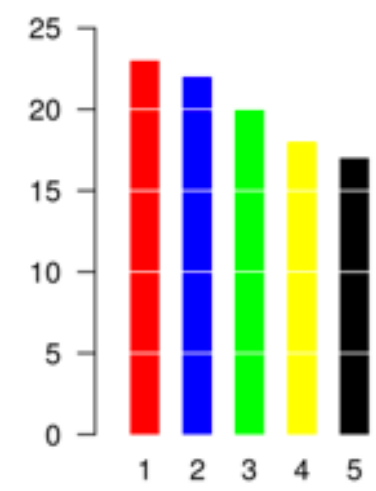
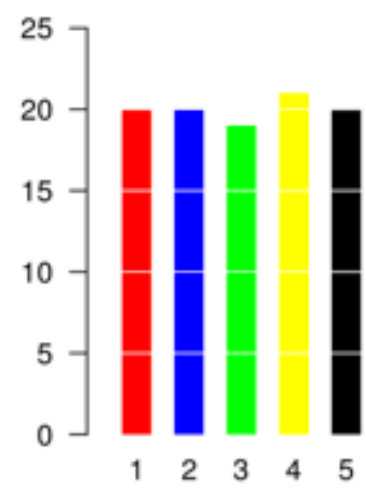
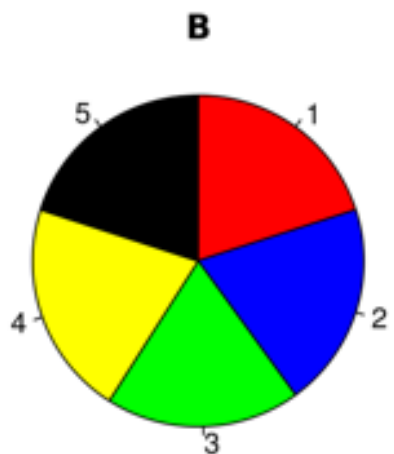
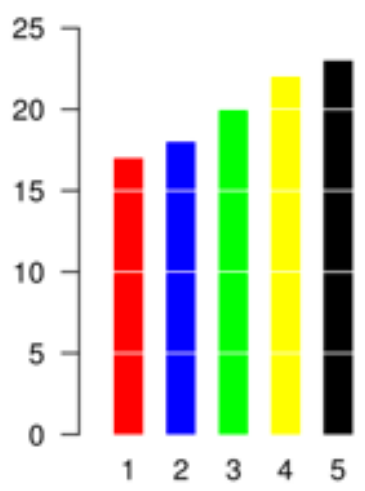
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Challenge: Find the biggest pie slice in each pie chart!

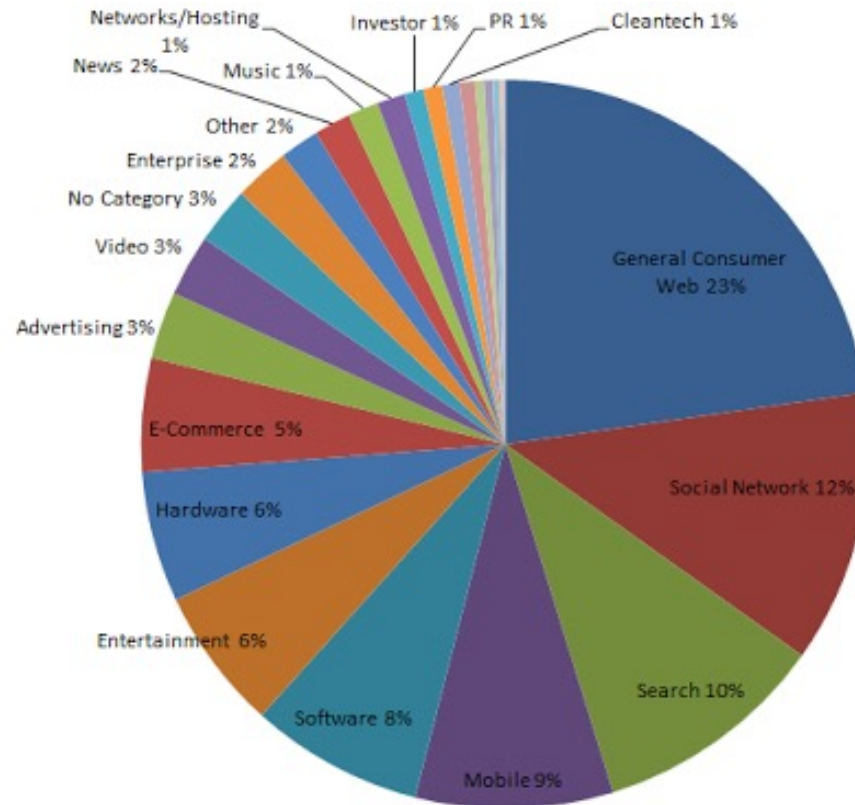


[Schutz 2007]



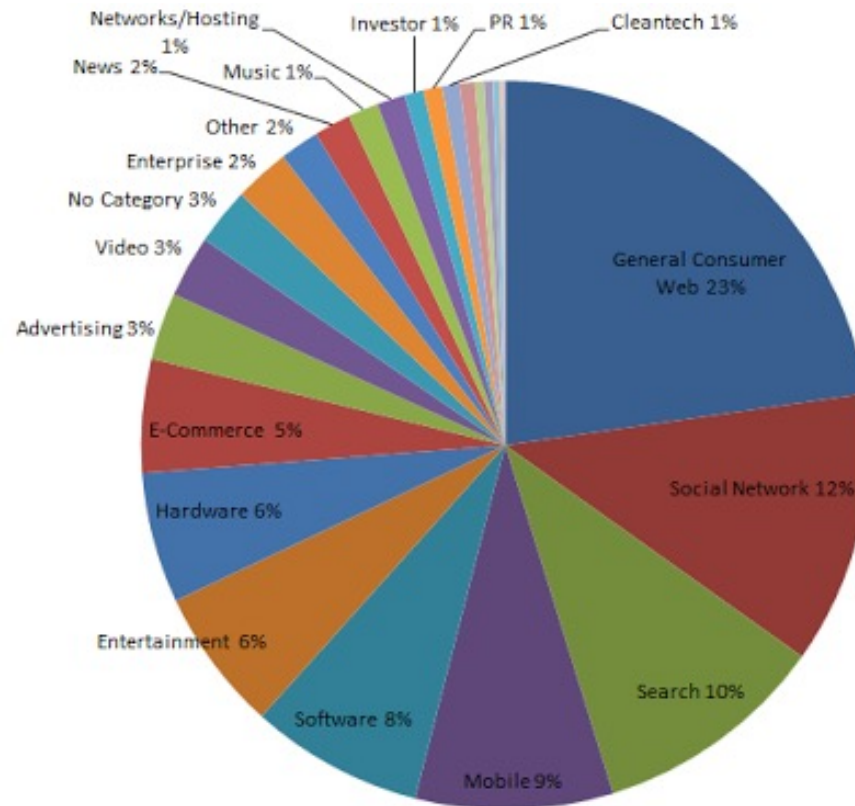
[Schutz 2007]

Share of coverage by topic on TechCrunch



[<http://www.storytellingwithdata.com/blog/2011/07/death-to-pie-charts>]

Share of coverage by topic on TechCrunch

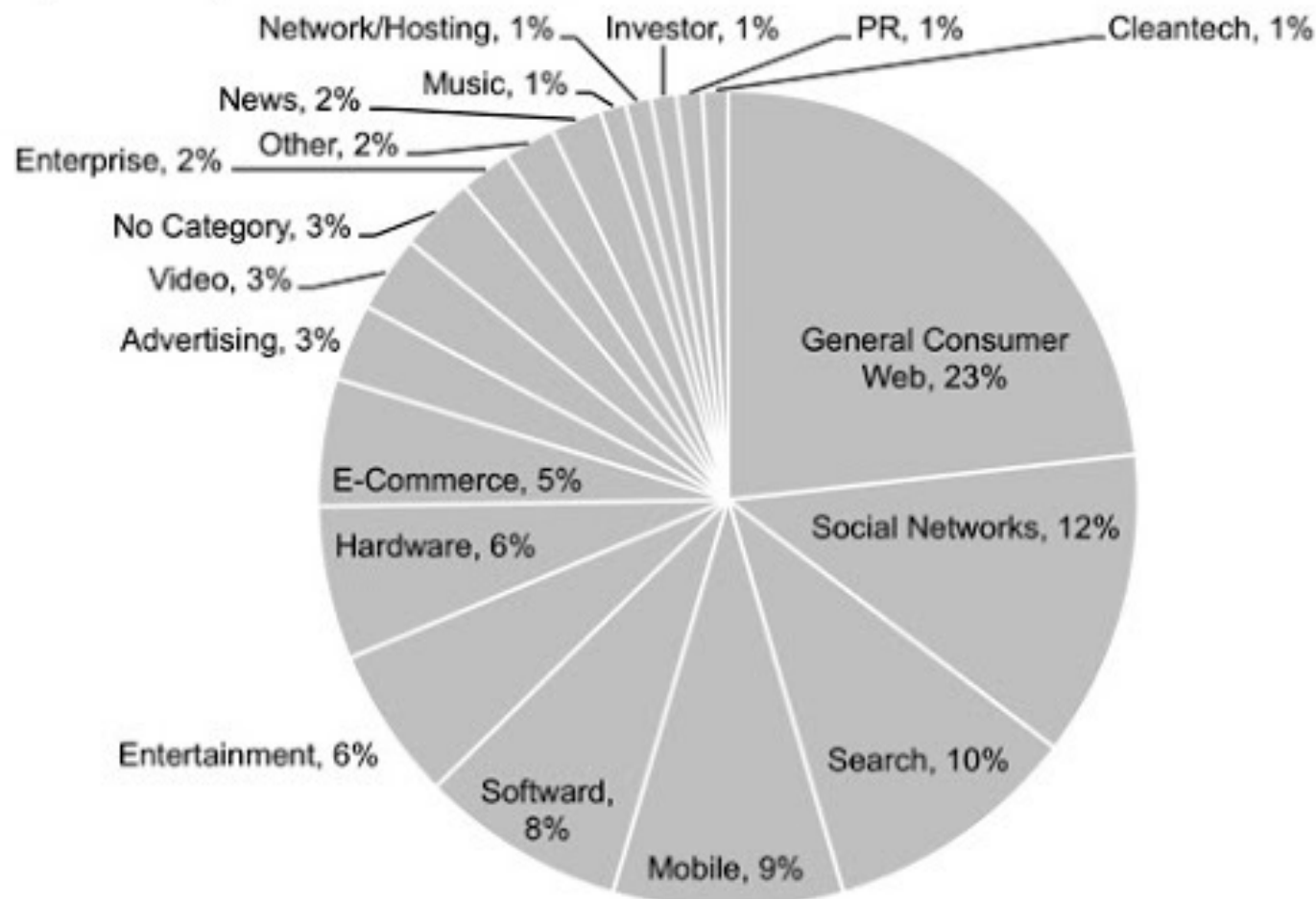


Too many slices!
Too many colors!
Bad color contrast!

[<http://www.storytellingwithdata.com/blog/2011/07/death-to-pie-charts>]

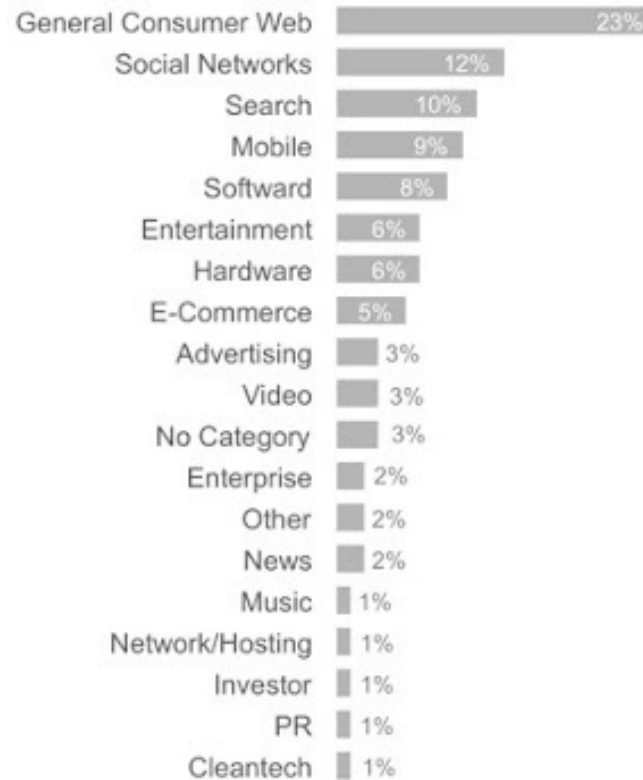
TechCrunch Coverage: 2005 - 2011

A slightly better pie?



TechCrunch Coverage: 2005 - 2011

Bars are best!



2012 PRESIDENTIAL RUN

GOP CANDIDATES



?!

FOX

47'

SOURCE: OPINIONS
DYNAMIC

2012 PRESIDENTIAL RUN

GOP CANDIDATES



SOURCE: OPINIONS
DYNAMIC

Math fail?!



World's Most Accurate Pie Chart

Some rules of thumb

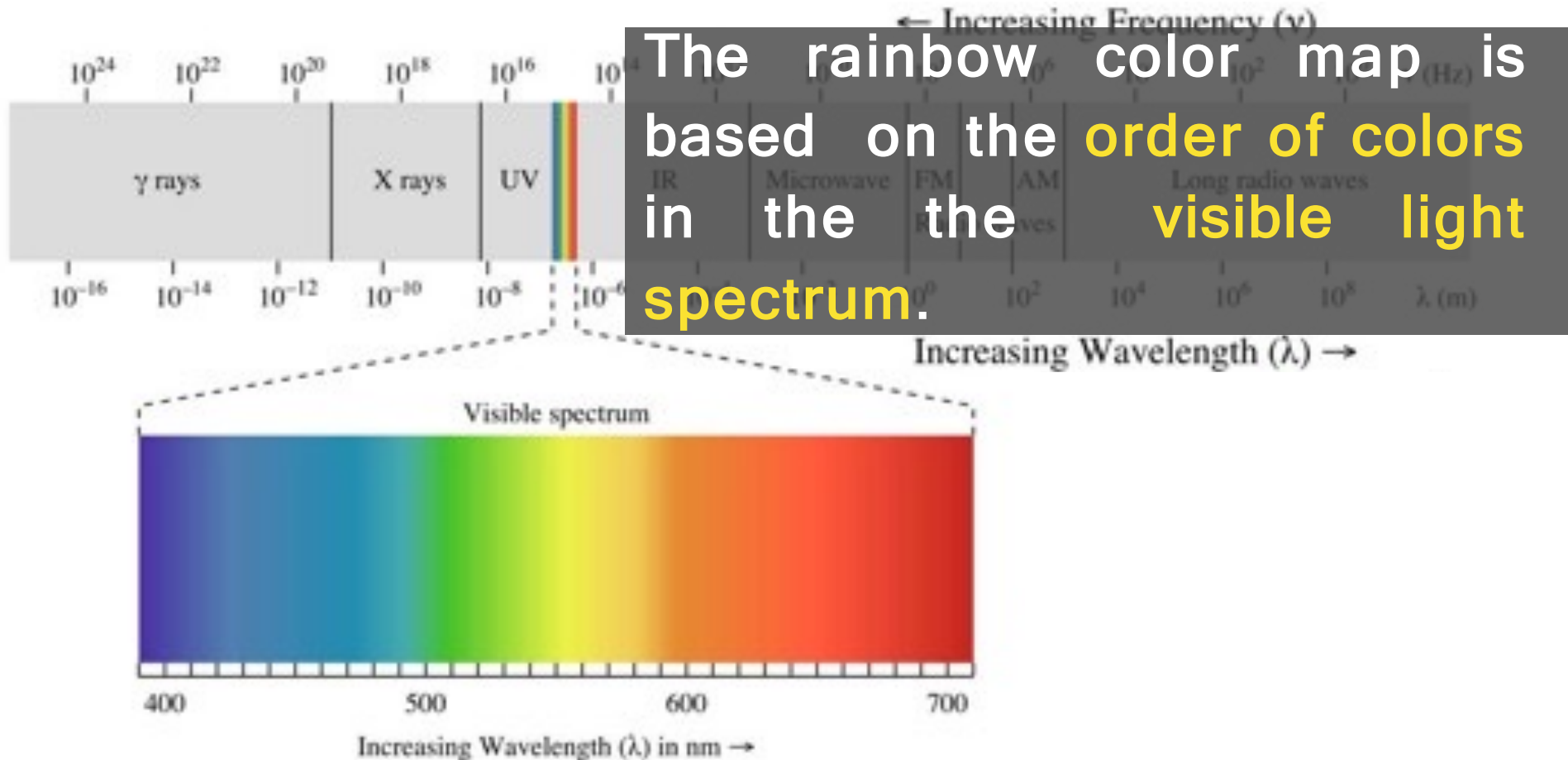
- You're comparing the parts to the whole.
- There's a small number of slices.
- Sort the values.
- Start at 12 o'clock.
- Use alternatives (e.g., bar chart)
- Pie charts are space efficient though

Rainbow Colormap



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<http://ractmytn.com/ractoids/visible-light-is-electromagnetic-radiation/>

SANFORD AND SELNICK

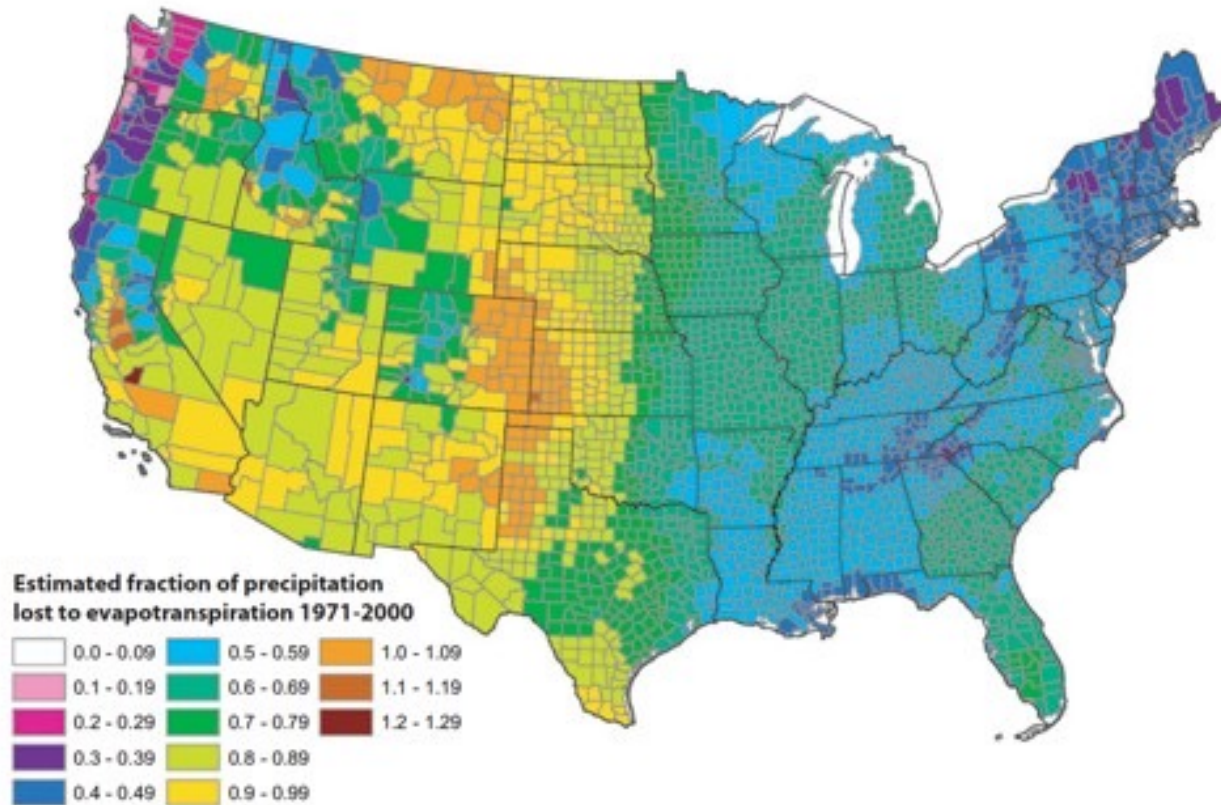
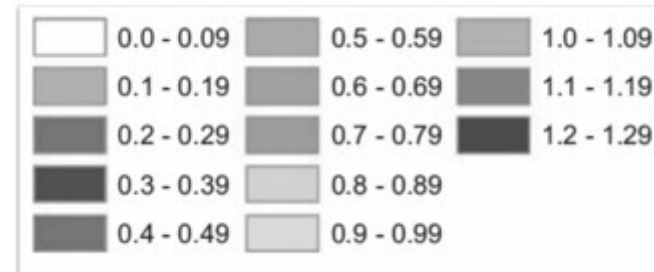
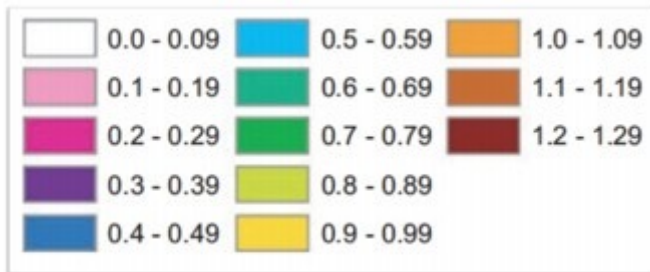


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

Can you say which **color** represents
a **higher** or **lower** value group?



Lack of perceptual ordering



[How The Rainbow Color Map Misleads \(Kosara 2013\)](#)

Some rules of thumb

Qualitative (rainbow) scheme – categorical data.



Sequential scheme – ordered categories or numerical data



Diverging scheme – numerical data with a meaningful mid-point.



[\[Color Brewer\]](#)

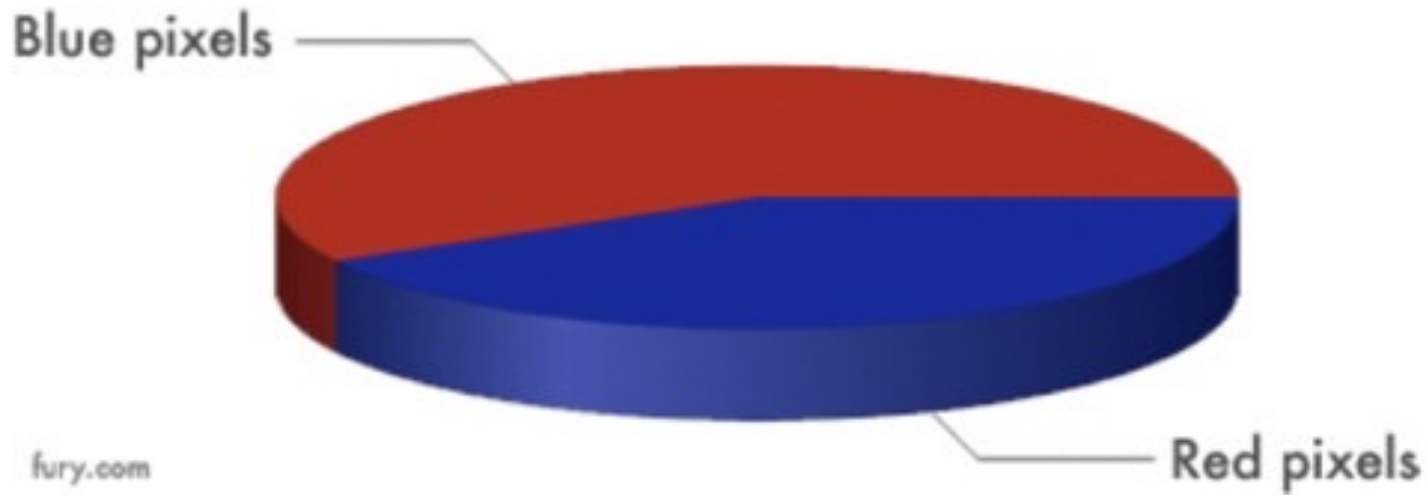
3D charts



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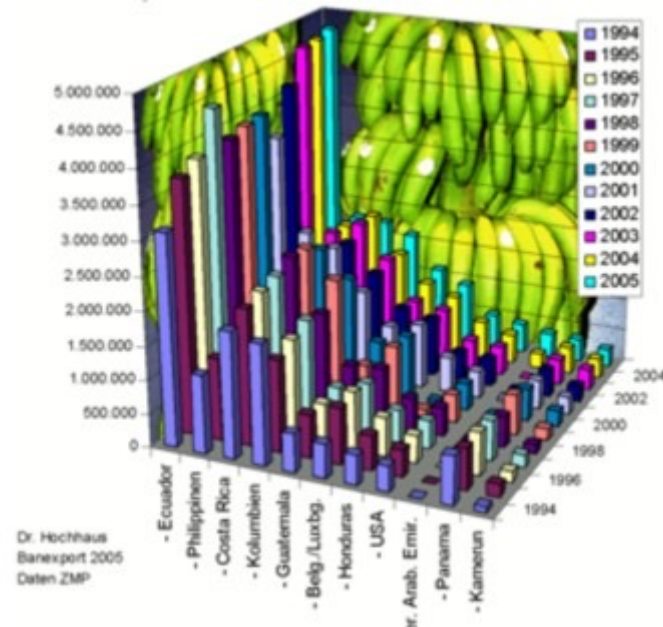
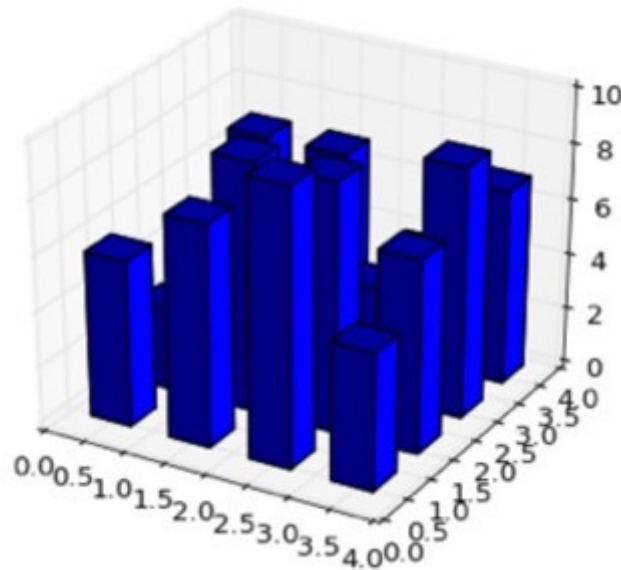
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Perspective distorts information

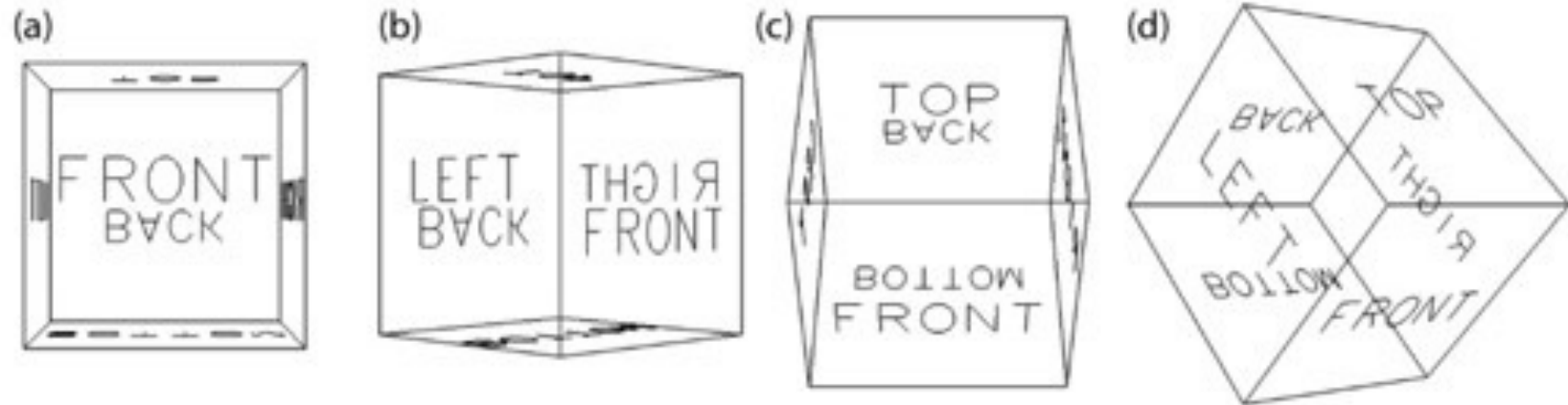


Occlusion hides information

Can rotate, but still no picture at once.

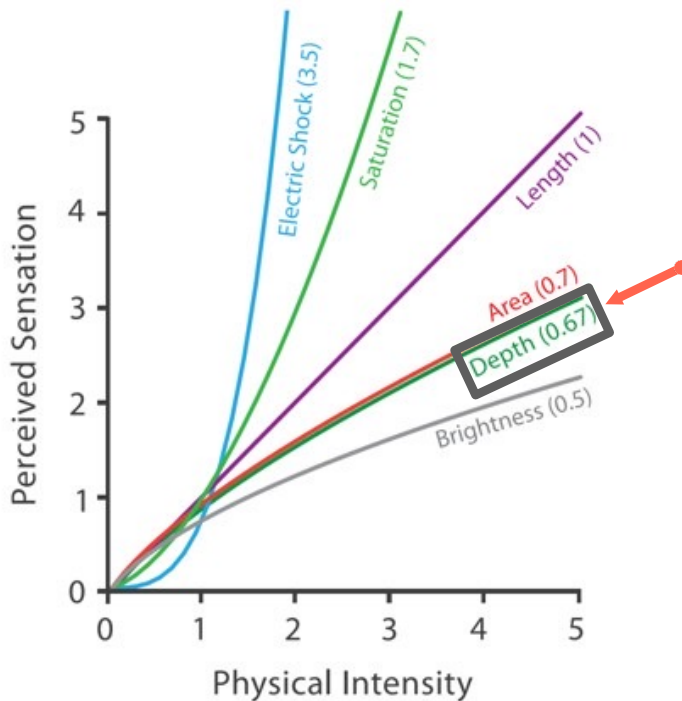


Tilted text isn't legible



Depth judgment is bad

Steven's Psychophysical Power Law: $S = I^N$



Human underestimate depth!

Actual intensity change

vs

Perceived Sensation

External cognition vs. Internal cognition



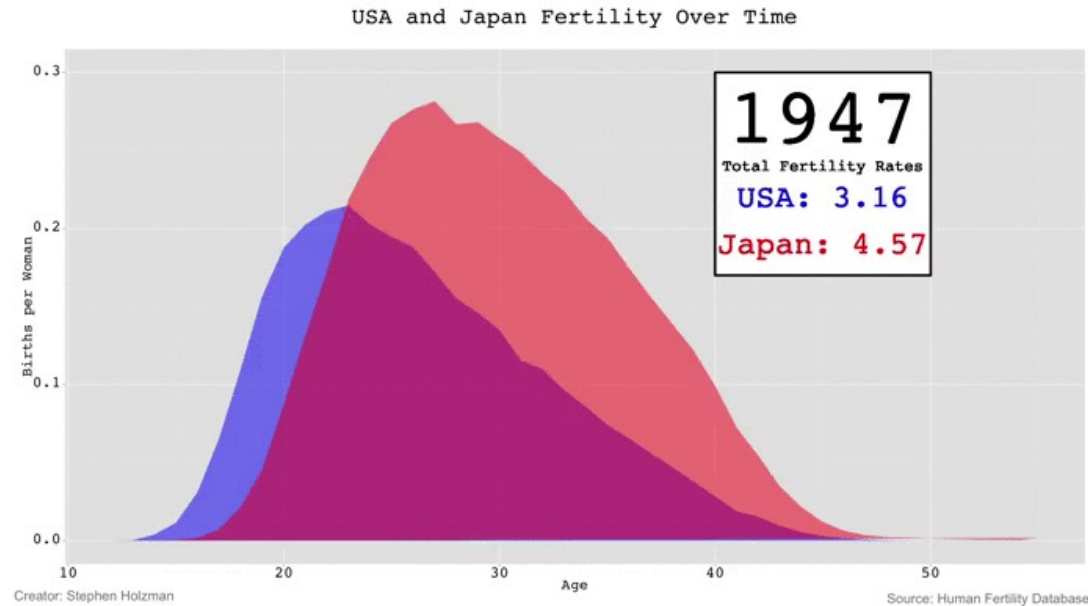
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Eyes beats memory

Animation

Harder to compare
visible item to
memory of what
you saw



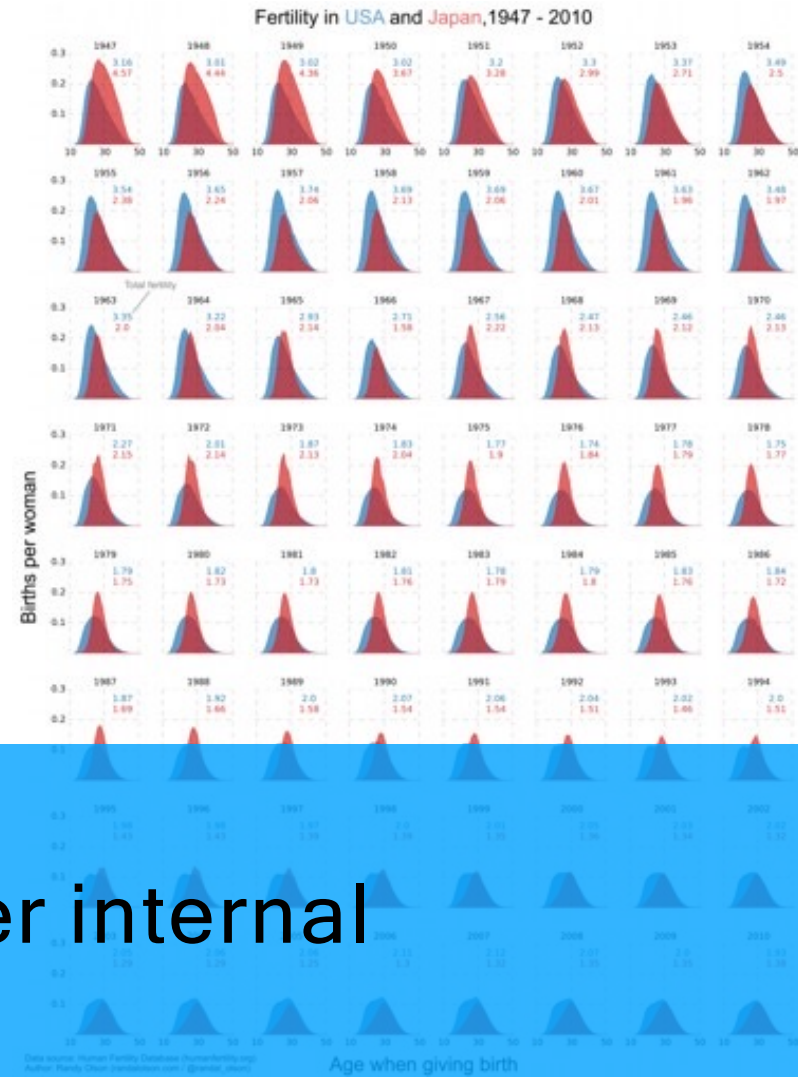
<http://www.randalolson.com/2015/08/23/small-multiples-vs-animated-gifs-for-showing-changes-in-fertility-rates-over-time/>

Eyes beats memory

Small multiples

Easy to compare by moving eyes
between side-by-side views

Use external cognition over internal
cognition



Subjective Dimensions

- Aesthetics: Attractive things are perceived as more useful.
- Style: Communicates brand, process, who the designer is.
- Playfulness: Encourages experimentation and exploration.
- Vividness: Can make a visualization more memorable.